

BSc Project Topics 2025-26

Themes:

Topics 1-25: Predictive Modelling

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Topics 261-275: Game Bots

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Theme: Predictive Modelling

#	Title	Description
1	Car Price Prediction using Supervised Learning	This project aims to develop a predictive model using machine learning algorithms to accurately estimate the market value of cars based on their features, such as make, model, age, mileage, and condition. [Data: kaggle.com/datasets/austinreese/craigslist-carstrucks-data]
2	Gold Price Forecasting using Regression Techniques	This project focuses on using historical gold price data and potentially related economic indicators to train a machine learning model that predicts future gold prices. [Data: kaggle.com/datasets/sid321axn/gold-price-prediction-dataset/data]
3	Flight Price Prediction using Machine Learning	This project focuses on using historical gold price data and potentially related economic indicators to train a machine learning model that predicts future gold prices. [Data: kaggle.com/datasets/sid321axn/gold-price-prediction-dataset/data]
4	Hotel Price Prediction using Regression Models	This project aims to build a machine learning model that predicts hotel prices based on key factors such as location, star rating, customer reviews, amenities, and seasonal demand patterns. The system will help identify how different features contribute to price variations, providing insights for both customers and hotel managers.[Data: kaggle.com/datasets/jiashenliu/515k-hotel-reviews-data-in-europe]
5	Real Estate Advertisement Price Prediction System	This project focuses on developing a predictive model that estimates real estate prices using property advertisement details, including location, size, number of rooms, property type, and additional features such as proximity to transport or schools. The model will be useful for property valuation, recommendation systems, and real estate market analysis. [Data: https://www.kaggle.com/datasets/fahmidaljaberprohor/bangladesh-real-estate-2025chittagong]

6	Insurance Premium Forecast using Linear Regression	The goal is to create a machine learning model that can predict the cost of insurance premiums for individuals based on their demographics, health information, and risk factors. [Data: kaggle.com/datasets/mirichoi0218/insurance/data]
7	House Price Forecasting using Machine Learning	In this project, time series models will be used for automated forecasting of house prices. The models will be trained and tested on public datasets. [Data: https://www.kaggle.com/datasets/yasserh/housing-prices-dataset]
8	Price Prediction for Coffee, Rice, and Beef Using Machine Learning	This project aims to develop predictive models to estimate price fluctuations for coffee, rice, and beef using historical agricultural data, trade statistics, and economic indicators. The models can help farmers, suppliers, and policymakers anticipate market shifts and make informed decisions across multiple commodities.[Data: https://www.kaggle.com/datasets/timmofey/-coffee-rice-and-beef-price-changes-for-30-years]
9	Churn Rate Prediction for Businesses	In this project, time series models will be used for automated forecasting of churning rates. The models will be trained and tested on public datasets. [Example dataset: https://www.kaggle.com/datasets/mnassrib/telecom-churn-datasets].
10	Customer Demand Forecasting	In this project, time series models will be used for automated forecasting of consumer demand. The models will be trained and tested on public datasets. [Data: https://www.kaggle.com/datasets/aswathrao/demand-forecasting]
11	Sports Team Performance Forecasting	In this project, time series models will be used for automated forecasting of the performance of sports teams. The models will be trained and tested on publicly available information of the performance of real teams. [Example data: https://www.kaggle.com/datasets/kaggleprollc/nba-forecast-data/data]
12	Stock Price Prediction	In this project, time series models will be used for automated forecasting of stock prices. The models will be trained and tested on public datasets. [Data: https://www.kaggle.com/datasets/luisandresgarcia/stock-market-prediction]
13	Weather Forecasting using Machine Learning	In this project, time series models will be used for automated forecasting of weather (e.g., temperature or humidity). The models will be trained and tested on public datasets.
14	Traffic Prediction using Historical Data	In this project, time series models will be used for automated forecasting of traffic intensity on a specific route at specific time, to assist driver in route planning and to optimise fuel consumption. The models will be trained and tested on public datasets.

15	Sleep Efficiency Prediction with EDA and ML	This project analyzes a sleep efficiency dataset by first cleaning and exploring the data. It uses visualizations to understand variable relationships, and feature engineering to enhance predictive power. Various machine learning models are then trained and evaluated to predict sleep efficiency. [Data: kaggle.com/code/melikedilekci/sleep-efficiency-dataset]
16	Airline Ticket Price Optimization with Dynamic Pricing	In this project, a computer program with two parts are developed for simulating consumer behaviour, and dynamically adjusting flight ticket prices to maximise profits. The program should be tested with different consumer behaviour configurations.
17	Hotel Dynamic Pricing Engine	In this project, a computer program with two parts are developed for simulating consumer behaviour, and dynamically adjusting hotel room prices to maximise profits. The program should be tested with different consumer behaviour configurations.
18	Train Ticket Dynamic Pricing	In this project, a computer program with two parts are developed for simulating consumer behaviour, and dynamically adjusting train ticket prices to maximise profits. The program should be tested with different consumer behaviour configurations.
19	Restaurant Dynamic Pricing System	In this project, a computer program with two parts are developed for simulating consumer behaviour, and dynamically adjusting restaurant prices to maximise profits. The program should be tested with different consumer behaviour configurations.
20	Customized Car Feature Pricing using Dynamic Pricing	In this project, a computer program with two parts are developed for simulating consumer behaviour, and dynamically adjusting the prices of different customised features (colour, transmission type, etc.) of cars to maximise profits. The program should take into consideration the customers' willingness to pay for the customisations, as well as the price of implementing them. The program should be tested with different consumer behaviour configurations.
21	E-commerce Sales Forecasting Using Seasonal Trends and ML	This project focuses on forecasting online sales by analyzing seasonal patterns, promotional effects, and consumer behavior trends. The model will help e-commerce businesses optimize inventory, pricing, and marketing strategies. [Data: kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data]
22	Energy Consumption Forecasting in Smart Homes	In this project, build a system that monitors energy consumption in commercial buildings and identifies opportunities for optimization, such as adjusting HVAC settings or scheduling energy-intensive tasks during off-peak hours.

23	Predicting Student Academic Performance using ML	This project develops a machine learning model to predict student academic outcomes based on features such as attendance, grades, demographics, and behavioral records. The model can support early intervention strategies for at-risk students.[Data: kaggle.com/datasets/aljarah/xAPI-Edu-Data]
24	Predictive Maintenance for Industrial Equipment	This project uses sensor readings, operational logs, and maintenance history to predict equipment failures before they occur. Such a model helps industries reduce downtime and maintenance costs while improving safety and efficiency. [Data: kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification]
25	Electricity Price Forecasting using Time-Series Models	This project aims to model and forecast electricity market prices using historical demand, generation, and trading data. Accurate predictions can support energy companies and policymakers in planning, trading, and stabilizing energy supply. [Data: kaggle.com/datasets/robikscube/hourly-energy-consumption]

Theme: Computer Vision

#	Title	Description
26	Dog Breed Classification	This projects develops machine learning models to accurately identify and classify different dog breeds. [Data: kaggle.com/datasets/mohamedchahed/dog-breeds]
27	Cat Breed Classification	This project aims to develop a deep learning model that can classify cat breeds from images using convolutional neural networks (CNNs). The model will help in pet identification, veterinary applications, and automated image tagging. data: https://www.kaggle.com/datasets/ma7555/cat-breeds-dataset
28	Fish Species Identification	This project focuses on developing a deep learning model to classify fish species from underwater images using Convolutional Neural Networks (CNNs). The model aims to assist marine biologists, fisheries management, and biodiversity studies by automating species identification in marine environments. [data: https://www.kaggle.com/datasets/sripaadsrinivasan/fish-species-image-data]
29	Street View House Number Detection	SVHN (Street View House Numbers): Contains real-world color images of house numbers collected from Google Street View images. The task is to recognize the digits in the images. [Source: kaggle.com/code/dimitriosroussis/svhn-classification-with-cnn-keras-96-acc/notebook]

30	Automated Register Plate Detection	This project aims to develop a computer vision system that can detect and read vehicle license plates from images and video using deep learning techniques. The system will be beneficial for applications such as traffic monitoring, parking management, and automated toll collection. [Data: https://www.kaggle.com/datasets/andrewmvd/car-plate-detection]
31	Supervised Learning for Strawberry Ripeness Classification	This is a precision agriculture project that leverages labeled image data to train a model capable of distinguishing between ripe and unripe strawberries, optimizing the harvesting process with high accuracy and efficiency. [Data: kaggle.com/datasets/abdulbasit31/strawberry-dataset]
32	Clustering-Based Feature Discovery for Strawberry Ripeness Detection	This project employs unsupervised clustering to discern patterns in strawberry images, such as color gradations and size variations, to unearth key features for ripeness classification, aiming for precise differentiation between ripe and unripe strawberries. [Data: kaggle.com/datasets/abdulbasit31/strawberry-dataset]
33	Unsupervised Machine Learning Based Analysis of Fruit Quality	The aim of the project is to train a machine learning classification model using descriptors such as size of the fruit, weight of the fruit, sweetness of the fruit, texture indicating crunchiness of the fruit etc., to predict quality of fruit in terms of good or bad. [Source: kaggle.com/datasets/nelgiriyeewithana/apple-quality ; kaggle.com/code/yossefmohammed/orange-quality-rating-prediction/notebook].
34	Tomato Leaf Detection by Transfer Learning	Using transfer learning, this project develops a model to detect and classify diseases in tomato leaves, aiming to aid early diagnosis and reduce crop loss for farmers. [Source: kaggle.com/code/adinishad/tomato-leaf-detection-by-transfer-learning/input]
35	Detect Retina Damage From OCT Images	This project aims to detect retina damage from Optical Coherence Tomography (OCT) images using machine learning or deep learning techniques. [Source: https://www.kaggle.com/code/paultimothymooney/detect-retina-damage-from-oct-images/notebook]
36	Retinal Vessel Segmentation Using Deep Learning	This project employs deep learning techniques to segment retinal blood vessels from images, aiding in medical diagnosis and analysis of various retinal diseases. [Source: https://www.kaggle.com/code/ipythonx/retinal-vessel-segmentation-starter/notebook]
37	Segmenting Northern European Glaciers from Satellite Imagery by Supervised Learning	Monitoring the melting of northern-European fresh-water glaciers is essential in understand the effects of global warming. This project aims to segment glaciers from satellite imagery by supervised machine learning methods. [Data: https://github.com/Aryal007/glacier_mapping]

38	Segmenting Northern European Glaciers Using Self-Supervised Learning and Few-Shot Transfer Learning	This project aims to segment glaciers from satellite imagery by machine learning methods with only a small number of human-annotated labels. Monitoring the melting of northern-European fresh-water glaciers is essential in understand the effects of global warming. [Data: https://github.com/Aryal007/glacier_mapping]
39	Classifying Crop Disease by Machine Learning	During this project you will design and develop machine learning methods for computer vision to detect the presence of disease in agricultural crops and further classify them into distinct disease categories. [Data: https://github.com/pratikkayal/PlantDoc-Dataset]
40	Identifying Disease Regions of Crops from Images by Machine Learning	During this project you will design and develop machine learning methods for computer vision to identify regions of disease in crop images to appropriately detect treatment areas. [Data: https://github.com/pratikkayal/PlantDoc-Dataset]
41	Plant Classification	During this project you will explore explainable AI methods such as GradCAM and SHAP (SHapley Additive exPlanations) to provide insights into the factors influencing a plant to be classified as diseased to improve identification and treatment decision making. [Data: https://github.com/pratikkayal/PlantDoc-Dataset]
42	Driving Scene Segmentation Using the Cityscapes Dataset	This project involves developing a deep learning model for semantic segmentation of urban driving scenes using the Cityscapes dataset. The goal is to classify each pixel into categories like roads, cars, and pedestrians, aiding autonomous driving systems in understanding and navigating complex urban environments. [Ref: kaggle.com/code/tr1gg3rtrash/car-driving-segmentation-unet-from-scratch]
43	Point Cloud Classification with Deep Learning	Implement a deep learning model for point cloud classification, and then train and test the model using publicly available point cloud datasets.
44	Image Super-Resolution Using Autoencoders	This project aims to develop a deep learning model to enhance the resolution of low-quality images using convolutional autoencoders. By training the model to learn the mapping from low-resolution to high-resolution images, the project seeks to improve image clarity and detail. Potential applications include medical imaging, satellite imagery, and photography enhancement.[data: https://www.kaggle.com/datasets/adityachandrasekhar/image-super-resolution]
45	Evaluation of Explainable AI Techniques for Image Classification	This project aims to assess different interpretability techniques for deep learning image classifiers, helping understand model decisions and improve trustworthiness. Applications include healthcare diagnostics and autonomous systems. [Data: kaggle.com/datasets/puneet6060/intel-image-classification]

46	MRI Image Denoising by Traditional Computer Vision	This project aims to improve the quality of MRI scans using traditional computer vision techniques such as filtering to facilitate better diagnosis and treatment planning for patients. [Data: https://brain-development.org/ixi-dataset/]
47	MRI Image Denoising by Machine and Deep Learning	This project aims to improve the quality of MRI scans using deep learning based computer vision techniques such as CNNs to facilitate better diagnosis and treatment planning for patients. [Data: https://brain-development.org/ixi-dataset/]
48	MRI Image Denoising by Hybrid Deep Learning and Filtering Methods	This project aims to improve the quality of MRI scans using a hybrid approach of traditional computer vision techniques such as filtering and deep learning methods to facilitate better diagnosis and treatment planning for patients. [Data: https://brain-development.org/ixi-dataset/]
49	Video Summarisation Using Deep Learning	This project will develop a basic video summarization tool that automatically selects key frames from a video to create a concise summary. It will also explore additional features such as frame importance scoring and a simple user interface to enhance usability. [Ref: kaggle.com/code/rushikeshhiray/rushikesh-hiray-vid-summary].
50	Image Quality Assessment	This project aims to develop machine learning models to evaluate the visual quality of images, focusing on sharpness, noise, and artifacts. The models will assist in applications such as photography, medical imaging, and content streaming platforms by automating the assessment of image quality.[data: https://www.kaggle.com/datasets/nisarahmedrana/biq2021]
51	Video Quality Assessment	This project aims to develop machine learning models to assess the visual quality of videos, focusing on factors such as blur, compression artifacts, and motion consistency. By extracting frame-level features and utilizing regression or deep learning techniques, the model will predict overall video quality. Applications include video streaming, broadcasting, and multimedia research. [Data : https://live.ece.utexas.edu/research/LIVEVQC/index.html]
52	Facial Expression Recognition	This project develops CNN models to detect and classify human facial expressions such as happiness, anger, and surprise. Students will explore pre-processing steps, data augmentation, and CNN architectures to improve accuracy. Applications include mental health monitoring, adaptive learning, and human-computer interaction. [Data: kaggle.com/datasets/msambare/fer2013]

53	Autonomous Vehicle Object Detection	This project focuses on developing deep learning models to detect vehicles in autonomous driving environments. By leveraging computer vision techniques, the model aims to enhance vehicle perception systems, contributing to safer and more efficient autonomous navigation. [Data: https://www.kaggle.com/datasets/abdallahwagih/cars-detection]
54	Counting People in Crowded Scenes	This project aims to develop machine learning models to estimate crowd sizes in public spaces by analyzing video feeds. The models will extract features such as density maps, motion patterns, and spatial distributions to predict the number of individuals present. Applications include event management, urban planning, and public safety monitoring. [data: https://www.kaggle.com/datasets/trainingdatapro/crowd-counting-dataset]
55	Face Mask Detection in Public Places	This project aims to develop a computer vision system that can detect whether individuals are wearing face masks in public areas. By leveraging deep learning techniques, the system will analyze images to classify individuals as wearing a mask, not wearing a mask, or wearing a mask incorrectly. This project can support health monitoring and compliance enforcement during pandemics.[data: https://www.kaggle.com/datasets/andrewmvd/face-mask-detection]
56	Road Sign Recognition for Autonomous Driving	This project focuses on classifying traffic signs using convolutional neural networks (CNNs) to support autonomous driving systems. Accurate recognition helps ensure vehicle safety and compliance with traffic rules. [Data: kaggle.com/datasets/valentynsichkar/traffic-signs-preprocessed]
57	Speed Limite Detection for Autonomous Driving	This project develops methods to automatically detect speed limits from traffic signs using deep learning-based vision systems. It can enhance safety and navigation for autonomous vehicles. [Data: https://www.kaggle.com/datasets/meowmeowmeowmeowmeow/gtsrb-german-traffic-sign]
58	Gesture Recognition Using Skeleton Tracking	This project uses skeletal tracking data and machine learning models to recognize human gestures in real time. Applications include gaming, human-computer interaction, and sign language interpretation. [Data: https://www.kaggle.com/datasets/imspارش/gesture-recognition]

59	Pedestrian Detection in Real-Time Video	This project focuses on detecting and tracking pedestrians in live video streams using object detection models like YOLO or SSD. Students will learn to handle real-time inference, video frame processing, and model evaluation. Applications include surveillance, urban safety, and autonomous vehicle navigation. [https://www.kaggle.com/datasets/karthika95/pedestrian-detection]
60	Surface Crack Detection in Buildings	This project develops a vision-based system to detect structural cracks in concrete or brick surfaces. Students will explore CNNs for image classification or segmentation and consider pre-processing techniques for crack enhancement. Applications include infrastructure inspection, maintenance, and safety monitoring. [Data: https://www.kaggle.com/datasets/oluwaseunad/concrete-and-pavement-crack-images]
61	Medical X-ray Image Classification	This project focuses on classifying X-ray images to detect diseases or abnormalities. By leveraging deep learning techniques, the model aims to assist healthcare professionals in diagnosing various conditions efficiently. [Data: kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia]
62	Dermatological Disease Classification	In this project, you'll design and develop machine learning methods for computer vision to detect skin diseases using dermatoscopic image datasets. The goal is to classify various skin conditions to aid in early detection and treatment. [Data: kaggle.com/datasets/andrewmvd/isis-2019]
63	Cell Counting in Microscopy Images	This project involves counting cells in microscopic images using segmentation techniques. The model will assist in biological research and medical diagnostics by automating cell counting processes. [Data: https://www.kaggle.com/datasets/gangadhar/nuclei-segmentation-in-microscope-cell-images]
64	Currency Recognition System	Develop a system to identify currency denominations from images using machine learning algorithms. This project aims to automate currency recognition for applications in finance and retail. [Data: kaggle.com/datasets/aishwaryatechie/usd-bill-classification-dataset]
65	Barcode and QR Code Detection and Decoding	Create a system to read and decode barcodes and QR codes from product packaging. The project will enhance inventory management and retail operations through automated code scanning. [Data: kaggle.com/datasets/jonathanimmanuel/barcode-and-qr]

66	Augmented Reality Marker Detection	This project focuses on detecting and tracking AR markers in images or video streams. Students will learn to overlay virtual content on real-world scenes using computer vision and deep learning. [Data: https://www.kaggle.com/datasets/crsuthikshnkumar/aruco-marker-data-set]
67	3D Face Reconstruction from 2D Images	This project involves reconstructing a 3D face structure from single or multiple 2D images using machine learning models. Students will explore techniques like graph neural networks and adversarial networks to generate realistic 3D face models, applicable in areas such as virtual reality, gaming, and biometric authentication. [Data: https://github.com/Juyong/3DFace]
68	Fire and Smoke Detection from Surveillance Cameras	Develop a system to identify early signs of fire or smoke in video feeds from surveillance cameras. By analyzing video frames, students will implement object detection and temporal analysis techniques to enhance safety monitoring in environments like forests, industrial sites, or urban areas. [Data: https://www.kaggle.com/datasets/ritupande/fire-detection-from-cctv]
69	License Plate Blur Detection for Privacy	This project focuses on detecting blurred license plates in images to support privacy protection. Students will work with image quality assessment models to identify and flag instances where license plates are intentionally or unintentionally obscured, ensuring compliance with privacy regulations. [Data: https://arxiv.org/html/2404.13677v1]
70	Multi-Label Image Classification	Design a model capable of classifying images with multiple simultaneous labels, such as identifying various objects within a single image. This project introduces students to multi-label classification techniques and their applications in areas like content tagging, medical imaging, and autonomous systems. [Data: https://www.kaggle.com/datasets/meherunnesashraboni/multi-label-image-classification-dataset]
71	Scene Text Recognition	Develop a system to detect and extract text from natural scene images, such as street signs or posters. Students will apply optical character recognition (OCR) and deep learning models to handle challenges like varying fonts, orientations, and lighting conditions in real-world scenarios. [Data: https://www.kaggle.com/datasets/ipythonx/totaltextstr]
72	Food Image Classification	Classify different food dishes from photographs using convolutional neural networks. This project is useful for dietary apps, menu recognition, and nutrition tracking. [Data: https://www.kaggle.com/datasets/kmader/food41]

73	Building Footprint Detection from Aerial Images	This project focuses on detecting and extracting building boundaries from high-resolution aerial or satellite images. Students will implement semantic segmentation or object detection models to generate accurate building footprints, supporting urban planning, disaster management, and geographic information systems (GIS). [Data: https://www.kaggle.com/datasets/joebeachcapital/australia-building-footprints]
74	Plastic Waste Detection in Ocean Images	This project focuses on detecting floating plastic debris in oceanic images, such as those captured by drones or satellites. Students will apply object detection and segmentation techniques to identify and classify plastic waste, contributing to environmental monitoring and conservation efforts. [data:https://www.kaggle.com/datasets/jocelyndumlao/seaclear-marine-debris-detection-and-segmentation]
75	Fashion Item Recognition for E-Commerce	This project involves identifying and classifying apparel and accessories from product images for online retail platforms. Students will implement deep learning-based classification or object detection models to improve search, recommendation, and inventory management in e-commerce applications. [Data: https://www.kaggle.com/datasets/paramaggarwal/fashion-product-images-dataset]
76	Colorization of Black and White Images	Convert grayscale images into realistic color images using deep learning models. Students will explore convolutional neural networks (CNNs) and generative models to understand colorization challenges, with applications in photo restoration, media, and entertainment. [Data: https://www.kaggle.com/datasets/shravankumar9892/image-colorization]
77	Skeleton-Based Human Action Recognition	Recognize human actions from videos using skeletal keypoint data, extracted with pose estimation models. Students will implement temporal models such as LSTMs or graph convolutional networks to classify actions, useful for sports analysis, surveillance, and human-computer interaction. [Data: https://www.kaggle.com/datasets/dasmehdixtr/berkeley-multimodal-human-action-database]
78	Vehicle Type Classification in Traffic Footage	Classify vehicles into categories like car, truck, bus, or motorcycle from traffic camera videos. Students will work with CNNs or video classification models to support traffic analysis, automated tolling, and urban mobility studies. [Data: https://www.kaggle.com/datasets/kaggleashwin/vehicle-type-recognition]

79	Indoor Navigation Using Visual Place Recognition	Develop a system to recognize specific indoor locations, such as rooms or corridors, using image-based visual place recognition. Students will implement feature extraction and image retrieval methods to assist robotics navigation or AR/VR applications. [Data: https://www.kaggle.com/datasets/alexandermelekhin/itlp-campus-indoor]
80	Text-to-Image Generation Evaluation	Assess the accuracy and realism of AI-generated images based on text prompts. Students will explore evaluation metrics, human-in-the-loop feedback, and model comparison techniques to improve generative AI quality for creative and industrial applications. [Data: https://www.kaggle.com/datasets/robikscube/textocr-text-extraction-from-images-dataset]
81	Sports Pose Estimation in Athlete Videos	This project focuses on tracking human joint positions from video footage to analyze athletic performance. Students will apply pose estimation models to extract keypoints and evaluate movement patterns, which can support coaching, injury prevention, and performance optimization. [Data: https://christianingwersen.github.io/SportsPose/]
82	Emotion Recognition from Cartoon Characters	Detect emotions from cartoon or animated characters using computer vision. Students will extract faces and train CNN-based classifiers (or fine-tune pretrained backbones) to predict expressions such as happy, angry, sad, and surprised, supporting animation analysis, sentiment studies, and interactive media. [Data: https://www.kaggle.com/datasets/somnath796/detect-emotions-of-your-favorite-toons]
83	Aerial Image Change Detection	This project aims to detect changes in aerial or satellite imagery over time, such as urban development or environmental changes. Students will learn techniques for image differencing, segmentation, and classification to identify areas of change efficiently. [Data: https://github.com/satellite-image-deep-learning/datasets]
84	Eye Gaze Tracking	Develop a system to estimate the direction of a user's gaze using camera images. Students will work with eye landmark detection and regression models to predict gaze points, applicable in human-computer interaction, accessibility, and AR/VR research. [Data: https://www.nature.com/articles/s41597-022-01200-0]
85	Pothole Detection in Road Images	This project focuses on detecting and classifying potholes from road images to support infrastructure monitoring. Students will apply object detection and segmentation models to identify damaged areas for automated reporting and maintenance planning. [Data: https://www.kaggle.com/datasets/sovitrath/road-pothole-images-for-pothole-detection]

86	3D Object Detection Using Stereo Vision	Develop models to detect objects in 3D using stereo camera images and depth information. Students will implement depth estimation and 3D bounding box detection to support robotics, autonomous navigation, or augmented reality.[Ref: https://openaccess.thecvf.com/content_CVPR_2020/papers/Peng_IDA-3D_Instance-Depth-Aware_3D_Object_Detection_From_Stereo_Vision_for_Autonomous_CVPR_2020_paper.pdf]
87	Instance Segmentation for Retail Shelf Products	This project aims to detect and segment individual products on retail shelves. Students will apply instance segmentation techniques like Mask R-CNN to identify, classify, and localize items for inventory management and automated checkout systems. [https://docs.ultralytics.com/datasets/detect/sku-110k/]
88	Scene Classification for Indoor vs Outdoor	Classify images as indoor or outdoor environments using computer vision methods. Students will explore feature extraction, CNNs, and transfer learning to distinguish scenes for applications in robotics, navigation, and AR/VR. [ref: https://www.kaggle.com/datasets/prithivsakthiur/indooroutdoornet-20k]
89	Style Transfer for Portrait Images	Apply artistic styles to portrait images using neural style transfer. Students will implement CNN-based models to transfer painting styles while preserving facial structure for creative and artistic applications.[Data: https://www.kaggle.com/datasets/deewakarchakraborty/portrait-paintings]
90	Counting Wildlife from Aerial Footage	Count animals from drone or aerial footage using object detection and segmentation models. Students will implement detection and counting algorithms to estimate wildlife populations in conservation areas. [Data: https://www.kaggle.com/datasets/kmldas/sea-turtles-in-drone-imagery]
91	Retail Checkout Object Detection	Detect and classify retail products at self-checkout counters using object detection models. Students will build pipelines for real-time item recognition, useful for automated retail systems. [Data: https://www.kaggle.com/datasets/diyer22/retail-product-checkout-dataset]
92	Human Activity Recognition from Video	Recognize human actions from videos using skeletal keypoint data or frame-based features. Students will implement temporal models such as LSTMs or graph convolutional networks to classify actions, useful for sports analysis, surveillance, and human-computer interaction. [Data: https://www.kaggle.com/datasets/sharjeelmazhar/human-activity-recognition-video-dataset]

93	Crowd Behavior Analysis	Analyze crowd movement patterns from video footage using detection, tracking, and density estimation techniques. Students will implement models for monitoring group behavior, useful for event safety and urban planning. [Data: https://www.kaggle.com/datasets/trainingdatapro/crowd-counting-dataset]
94	Nighttime Image Enhancement	Enhance low-light images using deep learning models for improved visibility. Students will experiment with CNNs or GANs to restore details in nighttime or dimly lit images for photography, surveillance, or autonomous vehicles. [Data: https://www.kaggle.com/datasets/washingtongold/exdark-dataset]
95	Rain Streak Removal in Images	Remove rain artifacts from outdoor images using CNN-based image restoration techniques. Students will build models to improve visual quality for photography, surveillance, and autonomous driving in rainy conditions. [Data: https://www.kaggle.com/datasets/ahmadahmadzada/images2000]
96	Animal Species Classification from Images	Classify animal species from photographs using deep learning models. Students will implement CNNs or transfer learning pipelines to distinguish between different animals, supporting applications in biodiversity monitoring, conservation, and automated image tagging. [Data: https://www.kaggle.com/datasets/iamsouravbanerjee/animal-image-dataset-90-different-animals]
97	Blood Cell Type Detection from Microscopic Images	Classify different blood cell types from microscope images to aid medical diagnostics. Students will learn segmentation and classification techniques for automated hematology analysis. [Data: https://www.kaggle.com/datasets/paultimothymooney/blood-cells]
98	Handwritten Signature Verification	Verify handwritten signatures using CNNs or Siamese networks to detect forgery. Students will implement feature extraction and classification pipelines for banking, legal, or authentication applications. [Data: https://www.kaggle.com/datasets/akashgundu/signature-verification-dataset]

99	Character Recognition in Historical Manuscripts	This project aims to develop a system for extracting and recognizing characters from scanned images of historical manuscripts using optical character recognition (OCR) techniques tailored for degraded or stylized texts. Students will preprocess images to handle challenges like faded ink, irregular fonts, or paper degradation, applying image processing methods such as binarization and noise reduction. Machine learning models, such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs), will be implemented using Python libraries like OpenCV, Tesseract, or PyTorch to recognize characters. The project offers practical experience in handling complex image datasets and contributes to preserving cultural heritage by digitizing historical texts. Students will evaluate model performance using metrics like character error rate and explore ethical considerations, such as ensuring accurate representation of historical content. Dataset: DIBCO (Document Image Binarization Contest) Dataset https://vc.ee.duth.gr/dibco2019/, Kaggle Historical Manuscripts Dataset https://www.kaggle.com/c/comp90051-2016
100	Augmented Reality Object Overlay	This project focuses on developing an augmented reality (AR) system that overlays digital objects onto real-world video feeds in real time, creating immersive AR experiences. Students will use computer vision techniques to detect and track real-world objects or markers in video streams, then superimpose 3D or 2D digital assets using tools like OpenCV, AR.js, or Unity. The project involves processing video inputs, implementing feature detection algorithms, and rendering digital objects seamlessly to align with the real-world environment. Students will gain hands-on experience with AR frameworks and explore applications in gaming, education, or visualization. The project also addresses challenges like occlusion handling and performance optimization, while considering ethical aspects such as user privacy and data security. Dataset: AR Marker Dataset (ArUco Markers) https://www.kaggle.com/datasets/crsuthikshnkumar/aruco-marker-dataset/data, NYU Depth Dataset V2 https://cs.nyu.edu/~silberman/datasets/nyu_depth_v2.html, COCO Dataset (Common Objects in Context) https://cocodataset.org/
101	Video Classification using 3D CNNs	Implement a 3D convolutional neural network and train it for a video classification task using publicly available video classification datasets.
102	Video Classification using CNN + RNN	Implement a deep model using CNN for image feature extraction and a recurrent neural network based on e.g., LSTM, for video action classification. Train and test the model using publicly available video action recognition datasets.

103	Image Deepfake Detection	This project involves creating algorithms to detect deepfake images produced by AI generative models, utilizing techniques like frequency domain analysis, artifact detection, and deep learning classifiers. Students will implement and train models using frameworks such as TensorFlow or PyTorch to differentiate between real and synthetic images based on subtle inconsistencies. The evaluation process will include benchmarking on diverse datasets, assessing metrics like accuracy, precision, recall, and robustness to variations in image quality. This endeavor provides practical experience in computer vision and AI ethics, highlighting the importance of combating digital misinformation. Students will also explore challenges such as adversarial robustness and the evolution of detection methods alongside advancing generation technologies. Dataset: CIFAKE: Real and AI-Generated Synthetic Images https://www.kaggle.com/datasets/birdy654/cifake-real-and-ai-generated-synthetic-images, Synthbuster Dataset https://zenodo.org/records/10066460/files/synthbuster.zip?download=1, WildFake Dataset https://github.com/hy-zpg/AIGC-Image-Detection-Dataset
104	Video Editing Automation Script	This project builds a script that automates basic video editing tasks like trimming, concatenating, adding watermarks, or fading in/out audio. It will use FFmpeg and Python as the core video and audio processing tool for multimedia tasks.
105	Video Deepfake Detection	This project focuses on constructing machine learning models to identify deepfake videos generated by AI, emphasizing the detection of anomalies in facial movements, audio-visual synchronization, and frame consistency. Students will utilize deep learning frameworks like PyTorch or TensorFlow to implement architectures such as 3D convolutional neural networks or recurrent models for video analysis. The evaluation phase involves testing models on benchmark datasets with metrics including accuracy, precision, recall, and robustness against various forgery techniques. Participants will gain experience in handling video data processing and addressing ethical implications of deepfakes in society. The project also explores strategies to improve model generalization and resistance to evolving AI generation methods. Dataset: SDFVD (Small-scale Deepfake Forgery Video Dataset) https://data.mendeley.com/datasets/bcmkfgct2s, Deepfake Detection Challenge (DFDC) Dataset https://ai.meta.com/datasets/dfdc/, ExDDV Dataset https://github.com/vladhondru25/ExDDV

Theme: Natural Language Processing

#	Title	Description
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106	Sentiment Analysis of Amazon Product Reviews	Identification of sentiment in the publicly available Kaggle Amazon product reviews dataset using deep learning methods.
107	Sentiment Analysis of Flipkart Product Reviews	Identification of sentiment in the publicly available Kaggle Flipkart product reviews dataset using deep learning methods.
108	Sentiment Analysis of Women's E-commerce Clothing Product Reviews	Identification of sentiment in the publicly available Kaggle women's e-commerce clothing product reviews dataset using deep learning methods.
109	Sentiment Analysis of Fine Food Product Reviews	Identification of sentiment in the publicly available Kaggle fine food product reviews dataset using deep learning methods.
110	Automatic Summarisation of TripAdvisor Restaurant Reviews (London)	The social platform TripAdvisor links users to restaurants by means of reviews posted by them. Using the information of these interactions, we can get valuable insights text summarisation. [Dataset available at https://www.kaggle.com/datasets/inigolopezrioboo/a-tripadvisor-dataset-for-nlp-tasks?select=London_reviews.csv]
111	Automatic Summarisation of TripAdvisor Restaurant Reviews (New York)	The social platform TripAdvisor links users to restaurants by means of reviews posted by them. Using the information of these interactions, we can get valuable insights text summarisation. [Dataset available at https://www.kaggle.com/datasets/inigolopezrioboo/a-tripadvisor-dataset-for-nlp-tasks?select=New_York_reviews.csv]
112	Automatic Summarisation of TripAdvisor Restaurant Reviews (Paris)	The social platform TripAdvisor links users to restaurants by means of reviews posted by them. Using the information of these interactions, we can get valuable insights text summarisation. [Dataset available at https://www.kaggle.com/datasets/inigolopezrioboo/a-tripadvisor-dataset-for-nlp-tasks?select=Paris_reviews.csv]
113	Emotion Recognition Using Transfer Learning on TripAdvisor Restaurant Reviews (London)	This project uses a publicly available GitHub dataset to train an emotion classifier. The classifier can be used to experiment with new datasets or add more emotions to the classification.
114	Emotion Recognition Using Transfer Learning on TripAdvisor Reviews (Barcelona)	This project uses a publicly available GitHub dataset to train an emotion classifier. The classifier can be used to experiment with new datasets or add more emotions to the classification.
115	Emotion Recognition Using Transfer Learning on TripAdvisor Reviews (New York)	This project uses a publicly available GitHub dataset to train an emotion classifier. The classifier can be used to experiment with new datasets or add more emotions to the classification.
116	Automatic Keyword Extraction from Harvard University Ratings and Reviews	Utilize the publicly available rich review texts for keyword extraction using deep learning methods.

117	Automatic Summarisation of Harvard University Ratings and Reviews	Understand the sentiment behind the textual reviews, distinguishing between academic and traveller feedback.
118	Sentiment and Emotion Analysis of Harvard University Ratings and Reviews	Understand the sentiment behind the textual reviews, distinguishing between academic and traveller feedback.
119	Automatic Keyword Extraction from Yelp Reviews	Utilize a Yelp reviews dataset that focuses on Domino's franchises in the US to test summarisation algorithms. [Dataset: https://www.kaggle.com/datasets/srees1988/user-reviews-for-nlp]
120	Automatic Summarisation of Yelp Reviews	Utilize a Yelp reviews dataset that focuses on Domino's franchises in the US to test automatic keyword extraction algorithms. [Data: https://www.kaggle.com/datasets/srees1988/user-reviews-for-nlp]
121	Sentiment and Emotion Analysis of Yelp Reviews	Identify the sentiment and emotional tone behind Yelp reviews that focus on Domino's franchises in the US. [Data: https://www.kaggle.com/datasets/srees1988/user-reviews-for-nlp]
122	Document Classification of Academic Papers	Test clustering algorithms capable of categorizing academic papers into relevant research domains or topics. [Data: https://www.kaggle.com/datasets/arplusman/papers-by-subject]
123	Automatic Indexing of Academic Papers	Test keyword extraction algorithms to identify themes present within the academic papers. [Data: https://www.kaggle.com/datasets/arplusman/papers-by-subject]
124	Automatic Keyword Extraction from Medium Articles	Utilize a Yelp reviews dataset that focuses on Domino's franchises in the US to test summarisation algorithms. Data: https://www.kaggle.com/datasets/srees1988/user-reviews-for-nlp]
125	Automatic Summarisation of Medium Articles	Utilize the text of data science Medium articles to test summarisation algorithms. Data: https://www.kaggle.com/datasets/meruvulikith/1300-towards-datascience-medium-articles-dataset
126	Emotion Analysis of Medium Articles	Understand the emotional tone behind data science Medium articles. [Data: https://www.kaggle.com/datasets/meruvulikith/1300-towards-datascience-medium-articles-dataset]
127	Dialogue System to Choose a Master's Course (Aberdeen)	A conversational agent intended to communicate with a human concerning MSc course selection.
128	Dialogue System to Choose an Undergraduate Course (Aberdeen)	A conversational agent intended to communicate with a human concerning postgraduate course selection.

129	Comparison of Review Recommendation Algorithms (Collaborative Filtering vs Influence Scoring)	Support business decision making by means of analysing and comparing two different recommendation algorithms in the context of social media. The dataset is publicly available and provided by the Max-Planck Institute.
130	Comparison of Review Recommendation Algorithms (Collaborative vs Content-Based Filtering)	Support business decision making by means of analysing and comparing two different recommendation algorithms in the context of social media. The dataset is publicly available and provided by the Max-Planck Institute.
131	The Use of Large Language Models for Decision Making	Fine-tuning large language models to assist in decision making on advertisement and shopping behaviour.
132	The Use of Large Language Models for Prediction	Use large language models for prediction of stock market, sales market, etc., using publicly available datasets.
133	Question Answering Tasks Using Public Datasets	Use of public datasets specifically created for question answering tasks and (for example, SquAD from Stanford).
134	Evaluation of Instruction Following Capability of LLMs	This project involves creating an evaluation system to test the ability of large language models (LLMs) to adhere to diverse user instructions, ranging from simple tasks to complex constraints like word counts or specific formats. Students will curate or use existing prompts, generate responses with open-source LLMs via libraries such as Hugging Face Transformers, and develop automated scripts to verify compliance using heuristics or metrics. The focus will be on quantifiable aspects, such as checking for required keywords, length adherence, or structural formats, to compute performance scores like success rates. This hands-on experience will deepen understanding of LLM alignment and limitations in real-world applications. Students will also consider ethical implications, including potential biases in instructions and the societal impact of improved instruction-following AI. Dataset: IFEval Dataset https://huggingface.co/datasets/google/IFEval, LLMBar Dataset https://github.com/princeton-nlp/LLMBar

135	Evaluation of Math Reasoning of LLMs	This project aims to assess the mathematical reasoning capabilities of large language models (LLMs) by evaluating their performance on arithmetic and algebraic problem-solving tasks. Students will design a suite of test prompts covering basic operations, word problems, and algebraic equations, using open-source LLMs via libraries like Hugging Face Transformers. The evaluation will measure accuracy, step-by-step reasoning quality, and error patterns, employing metrics such as exact match and partial credit scoring. The project provides hands-on experience in natural language processing and benchmarking, while exploring LLMs' limitations in mathematical logic. Students will also address ethical considerations, such as ensuring unbiased problem sets and the implications of deploying LLMs in educational contexts. Dataset: GSM8K Dataset (Grade School Math 8K) https://www.kaggle.com/datasets/thedevastator/grade-school-math-8k-q-a, MATH Dataset https://github.com/hendrycks/math, AQUA-RAT Dataset https://www.kaggle.com/datasets/dschettler8845/aqua-rat
136	Evaluation of Code Understanding of LLMs	This project involves assessing the code comprehension abilities of large language models (LLMs) through tasks such as code summarization, defect detection, and semantic similarity analysis. Students will curate or utilize existing benchmarks to input code snippets into LLMs using APIs or libraries like Hugging Face Transformers and analyze outputs for accuracy. Evaluation will incorporate metrics like semantic similarity scores, precision in bug identification, and qualitative assessments of explanations. The initiative offers practical insights into LLM strengths and weaknesses in software engineering contexts. Students will also examine ethical issues, including potential biases in code datasets and the reliability of AI in critical coding applications. Dataset: CodeXGLUE Dataset https://huggingface.co/datasets?search=code_x_glue, CodeSearchNet Dataset https://github.com/github/CodeSearchNet, CodeMMLU Dataset https://huggingface.co/datasets/Fsoft-AIC/CodeMMLU

137	Understanding Attention Layers in LLMs for QA Tasks	This project explores the role of attention layers in large language models (LLMs) by analyzing how individual attention heads influence performance on question-answering (QA) tasks through interpretability techniques. Students will use tools like Transformer Lens or BertViz to visualize attention patterns, identifying heads that specialize in capturing relationships between questions and context passages. Experiments will involve probing models such as BERT or Llama on QA benchmarks, measuring changes in metrics like exact match and F1-score when specific heads are masked or pruned. The study will provide insights into model internals, helping to understand why certain heads are crucial for tasks like entity linking or inference. Students will also discuss implications for model efficiency and robustness, promoting advancements in interpretable AI. Dataset: SQuAD (Stanford Question Answering Dataset) https://rajpurkar.github.io/SQuAD-explorer/, HotpotQA:https://github.com/hotpotqa/hotpot
138	Understanding MLP Layers in LLMs for Dialogue Systems	This project delves into the contributions of multi-layer perceptron (MLP) layers within large language models (LLMs) to their effectiveness in dialogue systems, using interpretability techniques to uncover layer-specific functions. Students will analyze MLP activations and weights during dialogue processing, employing methods like activation patching or visualization to determine their role in maintaining coherence, context, and response generation. By conducting experiments on dialogue datasets, participants will measure performance impacts through metrics such as BLEU scores, perplexity, and human-likeness when modifying MLP components. The study aims to reveal how MLPs handle tasks like intent recognition and emotional nuance in conversations. Students will also consider ethical implications, including the interpretability of AI in interactive systems and potential biases amplified by these layers. Dataset: DialogBench Dataset https://github.com/kwai/DialogBench, DailyDialog Dataset https://www.kaggle.com/datasets/thedevastator/dailydialog-multi-turn-dialog-with-intention-and

139	Evaluation of Multilingual Understanding in LLMs	This project focuses on evaluating the ability of large language models (LLMs) to process and respond accurately to inputs across multiple languages, assessing tasks like translation, comprehension, and context retention. Students will design experiments using multilingual prompts, leveraging open-source LLMs through libraries like Hugging Face Transformers to generate responses. Performance will be measured with metrics such as BLEU, accuracy, and semantic correctness across languages with varying scripts and structures. The project provides practical experience in natural language processing and cross-lingual model evaluation, highlighting LLM limitations in low-resource languages. Students will also explore ethical considerations, including cultural biases and equitable model performance across diverse linguistic communities. Dataset: Flores-200 Dataset https://huggingface.co/datasets/facebook/flores, XTREME Dataset https://github.com/google-research/xtreme
140	Evaluation of Hallucinations in LLMs for Fact Checking	This project investigates hallucinations in large language models (LLMs), where models generate plausible but factually incorrect information, with a focus on fact-checking applications. Students will use benchmarks to prompt LLMs with factual queries, analyzing outputs for inaccuracies by comparing them to verified ground truths using metrics like factual consistency scores. The evaluation will categorize hallucination types, such as intrinsic (contradicting input) or extrinsic (unverifiable claims), and explore factors influencing them, like model size or prompting strategies. Practical experience will be gained in implementing detection methods, such as uncertainty estimation or reference-based checking, via libraries like Hugging Face Transformers. Students will also address ethical concerns, including the risks of deploying hallucination-prone LLMs in information-sensitive domains like journalism or education. Dataset: HaluEval Dataset https://github.com/RUCAIBox/HaluEval, DefAn Dataset https://github.com/ashikiut/DefAn

141	Sentiment Analysis of COVID-19 Kaggle Dataset Using SVM	This project focuses on performing sentiment analysis on COVID-19-related tweets or comments from a Kaggle dataset, using Support Vector Machines (SVM) to classify text as positive, negative, or neutral. Students will preprocess the dataset by cleaning text, removing noise like URLs and emojis, and extracting features using techniques such as TF-IDF or word embeddings. The SVM model, implemented via Python libraries like scikit-learn, will be trained to predict sentiment, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. This project provides hands-on experience in natural language processing and machine learning, offering insights into public sentiment during the pandemic. Students will also address ethical considerations, such as handling biased or sensitive social media content responsibly. Dataset: COVID-19 Tweets Dataset (Kaggle) https://www.kaggle.com/datasets/gpreda/covid19-tweets, COVID-19 Twitter Sentiment Dataset (Kaggle) https://www.kaggle.com/datasets/datatattle/covid-19-nlp-text-classification, Tweet Sentiment Extraction Dataset (Kaggle) https://www.kaggle.com/competitions/tweet-sentiment-extraction/data
142	Sentiment Analysis of COVID-19 Kaggle Dataset Using Llama 2	This project leverages Llama 2, a powerful large language model, to perform sentiment analysis on COVID-19-related text data, such as tweets or comments, classifying them as positive, negative, or neutral. Students will preprocess the dataset by cleaning text, handling noise like emojis or URLs, and preparing prompts for Llama 2 using libraries like Hugging Face Transformers. The analysis will involve fine-tuning or zero-shot prompting Llama 2 to predict sentiment, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project offers hands-on experience in applying advanced LLMs to real-world NLP tasks and provides insights into public sentiment during the pandemic. Students will also address ethical considerations, such as mitigating biases in social media data and ensuring responsible use of AI outputs. Dataset: COVID-19 Tweets Dataset (Kaggle) https://www.kaggle.com/datasets/gpreda/covid19-tweets, COVID-19 Twitter Sentiment Dataset (Kaggle) https://www.kaggle.com/datasets/datatattle/covid-19-nlp-text-classification, Tweet Sentiment Extraction Dataset (Kaggle) https://www.kaggle.com/competitions/tweet-sentiment-extraction/data

143	Hate Speech Detection on Twitter	This project aims to develop a natural language processing (NLP) system to identify offensive or hateful speech in Twitter data, leveraging machine learning models such as BERT or traditional classifiers like SVM for text classification. Students will preprocess tweets by cleaning text, removing noise like URLs and mentions, and extracting features using techniques like TF-IDF or transformer embeddings. The system will be trained to classify tweets as hateful, offensive, or neutral, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. This project provides hands-on experience in NLP and text classification while addressing real-world issues of online toxicity. Students will also explore ethical considerations, including mitigating biases in training data and ensuring fair detection across diverse user groups. Dataset: Twitter Sentiment Analysis Dataset (Kaggle) https://www.kaggle.com/datasets/kazanova/sentiment140, Hate Speech and Offensive Language Dataset (Kaggle) https://www.kaggle.com/datasets/mrmorj/hate-speech-and-offensive-language-dataset
144	Toxic Comment Classification in Online Communities	This project focuses on developing a natural language processing (NLP) system to classify toxicity levels in comments from online communities, categorizing them as toxic, non-toxic, or varying degrees of severity using machine learning models like BERT or logistic regression. Students will preprocess comment data by cleaning text, removing noise such as special characters, and extracting features using techniques like TF-IDF or transformer embeddings. The system will be trained to detect toxic language, with performance evaluated using metrics such as accuracy, precision, recall, and F1-score. The project provides practical experience in NLP and text classification while addressing the societal challenge of online harassment. Students will also explore ethical issues, including mitigating biases in training data and ensuring equitable detection across diverse communities. Dataset: Toxic Comment Classification Challenge Dataset (Kaggle) https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge/data, Jigsaw Unintended Bias in Toxicity Classification Dataset (Kaggle) https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification/data
145	Fake News Detection with NLP	In this project, a program is developed to automatically detect fake news, based on public datasets.

146	Detecting Clickbait in News Headlines	This project aims to develop a system to detect clickbait in news headlines by combining keyword pattern analysis with advanced natural language processing (NLP) models, such as BERT or logistic regression. Students will preprocess headline data, identifying sensationalist phrases and extracting features like TF-IDF or transformer embeddings. The system will classify headlines as clickbait or non-clickbait, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and text classification while addressing the societal issue of misleading information in media. Students will also explore ethical considerations, such as avoiding biases in classification and ensuring transparency in labeling potentially subjective content. Dataset: Clickbait Dataset (Kaggle) https://www.kaggle.com/datasets/amananandrai/clickbait-dataset
147	Legal Document Classification	This project focuses on developing a natural language processing (NLP) system to automatically classify legal documents, such as contracts, court rulings, or statutes, into predefined categories or topics using machine learning models like BERT or traditional classifiers like SVM. Students will preprocess text data by cleaning documents, handling legal jargon, and extracting features using techniques such as TF-IDF or transformer embeddings. The system will be trained to categorize documents accurately, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project offers practical experience in NLP and text classification while addressing real-world applications in legal tech, such as document management and retrieval. Students will also explore ethical considerations, including ensuring model fairness and handling sensitive legal information responsibly. Dataset: Legal Document Dataset (Kaggle) https://www.kaggle.com/datasets/amohankumar/legal-text-classification-dataset, CaseLaw Access Project (CAP) Dataset https://case.law/

148	Named Entity Recognition for Biomedical Texts	This project focuses on developing a named entity recognition (NER) system to identify and classify entities such as genes, diseases, and drugs within biomedical texts using natural language processing (NLP) techniques, such as transformer-based models like BERT or traditional methods like CRF. Students will preprocess biomedical literature by tokenizing text and annotating entities, leveraging libraries like spaCy, Hugging Face Transformers, or NLTK for model implementation. The system will be trained to accurately extract entities, with performance evaluated using metrics like precision, recall, and F1-score. The project provides hands-on experience in NLP and biomedical text mining, contributing to applications in medical research and drug discovery. Students will also address ethical considerations, such as ensuring model accuracy to avoid misidentification in critical healthcare contexts and handling sensitive data responsibly. Dataset: NCBI Disease Corpus https://www.ncbi.nlm.nih.gov/research/bionlp/Data/ , i2b2/VA 2010 Dataset https://www.i2b2.org/NLP/DataSets/
149	Machine Translation System for Low-Resource Languages	This project focuses on creating a machine translation (MT) system for low-resource languages that lack abundant parallel data, employing strategies like transfer learning from high-resource languages or synthetic data generation. Students will choose specific low-resource language pairs, preprocess available corpora to handle noise and scarcity, and implement neural models such as Transformers using Python libraries like Hugging Face Transformers or Fairseq. The system will incorporate techniques like back-translation or multilingual pre-training to enhance translation accuracy despite limited resources. Performance will be assessed using standard metrics such as BLEU, METEOR, or chrF, along with qualitative evaluations for fluency and adequacy. Students will also examine ethical implications, including the preservation of linguistic diversity and mitigation of cultural biases in automated translations. Dataset: FLORES-200 Dataset https://huggingface.co/datasets/facebook/flores, Masakhane MT Dataset https://github.com/masakhane-io/masakhane-mt

150	Abstractive Summarisation of Research Articles	This project entails utilizing generative NLP models, such as BART or T5, to produce abstractive summaries of extensive research articles, distilling essential elements like objectives, methodologies, results, and conclusions into concise narratives. Students will preprocess scientific texts to manage long sequences and specialized vocabulary, employing techniques like chunking or hierarchical encoding for effective input handling. The models will be fine-tuned on domain-specific datasets to enhance summary coherence and factual accuracy, incorporating methods to minimize hallucinations. Performance assessment will involve quantitative metrics such as ROUGE, METEOR, and BERTScore, alongside qualitative reviews for relevance and readability. Students will also delve into ethical issues, including the prevention of misinformation through faithful summarization and the promotion of accessible scientific knowledge. Dataset: scientific_papers Dataset (Hugging Face) https://huggingface.co/datasets/armanc/scientific_papers, PubMed Summarization Dataset (Hugging Face) https://huggingface.co/datasets/ccdv/pubmed-summarization, PubMed Article Summarization Dataset (Kaggle) https://www.kaggle.com/datasets/thedevastator/pubmed-article-summarization-dataset
151	Grammar Error Detection and Correction	This project involves building a natural language processing (NLP) system to detect grammatical errors in user-provided text and suggest corrections, utilizing models like transformers for sequence-to-sequence tasks. Students will preprocess input text to handle variations in writing styles and error types, implementing techniques such as tokenization and error tagging. The system will leverage fine-tuned generative models, such as T5 or BART, to generate corrected versions of the input while preserving original meaning. Evaluation will focus on metrics like GLEU score, precision in error detection, and overall fluency of corrections. Students will also consider ethical aspects, including avoiding over-correction that alters intent and ensuring accessibility for non-native speakers. Dataset: JFLEG Dataset https://huggingface.co/datasets/jhu-clsp/jfleg, CoNLL-2014 Shared Task Dataset https://www.comp.nus.edu.sg/~nlp/conll14st.html, BEA-2019 Shared Task Dataset https://www.cl.cam.ac.uk/research/nl/bea2019st/

152	Relation Extraction from Scientific Literature	This project focuses on developing a natural language processing (NLP) system to extract scientific relationships, such as protein-protein interactions or drug-disease associations, from scientific literature using advanced models like BERT or BioBERT. Students will preprocess text from research articles, handling complex scientific terminology and sentence structures through tokenization and entity annotation. The system will be trained to identify and classify relationships between entities, with performance evaluated using metrics like precision, recall, and F1-score. The project provides hands-on experience in NLP and relation extraction, contributing to applications in biomedical research and knowledge graph construction. Students will also explore ethical considerations, ensuring accurate extraction to avoid misinterpretation in critical scientific contexts and managing sensitive data responsibly. Dataset: BioNLP Shared Task Datasets (BioNLP-ST) https://2011.bionlp-st.org/, ChemProt Dataset https://huggingface.co/datasets/bigbio/chemprot, SciERC Dataset https://huggingface.co/datasets/hrithikpiyush/scierc
153	Resume Screening System Using NLP	This project aims to develop a natural language processing (NLP) system to automatically screen resumes by identifying and ranking them based on relevant job-specific keywords and skills. Students will preprocess resume texts, extracting features like job titles, skills, and experience using techniques such as TF-IDF, word embeddings, or transformer models like BERT. The system will employ classification or ranking models, implemented via libraries like scikit-learn or Hugging Face Transformers, to match resumes to job descriptions. Performance will be evaluated using metrics like precision, recall, and F1-score for keyword matching and ranking accuracy. Students will also address ethical considerations, ensuring fairness in screening to avoid biases related to gender, ethnicity, or other protected attributes. Dataset: Resume Dataset (Kaggle) https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset, Job Description and Resume Matching Dataset (Kaggle) https://www.kaggle.com/datasets/shamimhasan8/resume-vs-job-description-matching-dataset

154	Automatic Headline Generation for News Articles	This project focuses on developing a natural language processing (NLP) system to generate concise and relevant headlines from full-length news articles using generative models like T5, BART, or GPT-based architectures. Students will preprocess article texts to handle varying lengths and styles, employing techniques like text truncation or key sentence extraction to prepare inputs for the model. The system will be trained or fine-tuned to produce accurate and engaging headlines, with performance evaluated using metrics such as ROUGE, BLEU, and human-judged relevance. The project offers hands-on experience in NLP and text generation, contributing to applications in automated journalism and content curation. Students will also address ethical considerations, including avoiding sensationalism and ensuring factual fidelity in generated headlines. Dataset: All the News Dataset (Kaggle) https://www.kaggle.com/datasets/cymerunaslam/allthenews, CNN/Daily Mail Dataset https://huggingface.co/datasets/abisee/cnn_dailymail
155	Topic Modeling of Social Media Posts	This project focuses on applying topic modeling techniques, such as Latent Dirichlet Allocation (LDA) or BERTopic, to uncover latent themes and trends in social media posts, enabling insights into user discussions and sentiments. Students will preprocess text data from platforms like Twitter or Reddit, cleaning noise such as hashtags, URLs, and emojis, and preparing inputs using libraries like scikit-learn for LDA or Hugging Face for BERTopic. The system will extract coherent topics, visualized through word clouds or topic distributions, and performance will be evaluated using metrics like coherence scores (e.g., C_v, UMass). The project provides hands-on experience in natural language processing and unsupervised learning, applicable to areas like market analysis and public opinion tracking. Students will also address ethical considerations, such as handling biased or sensitive content and ensuring privacy in social media data analysis. Dataset: Twitter Sentiment Analysis Dataset (Kaggle) https://www.kaggle.com/datasets/kazanova/sentiment140, Reddit Comments Dataset (Kaggle) https://www.kaggle.com/datasets/smagnan/1-million-reddit-comments-from-40-subreddits,

156	Text Classification for Mental Health Indicators	This project aims to develop a natural language processing (NLP) system to classify social media text for indicators of mental health issues, such as anxiety, depression, or stress, using machine learning models like BERT or traditional classifiers like SVM. Students will preprocess text data from platforms like Twitter or Reddit, cleaning noise such as emojis, hashtags, and URLs, and extracting features using TF-IDF or transformer embeddings. The system will be trained to identify linguistic patterns associated with mental health conditions, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and text classification while contributing to mental health awareness through early detection. Students will also address ethical considerations, including ensuring user privacy, avoiding stigmatization, and handling sensitive data responsibly. Dataset: Mental Health Dataset (Kaggle) https://www.kaggle.com/datasets/bhavikjikadara/mental-health-dataset, CLPsych Shared Task Datasets https://clpsych.org/shared-task-2024/
157	Aspect-Based Sentiment Analysis for Product Reviews	This project aims to develop a natural language processing (NLP) system for aspect-based sentiment analysis (ABSA), identifying and classifying sentiments toward specific product features (e.g., battery life, camera quality) in customer reviews. Students will preprocess review texts, extracting aspects and sentiments using techniques like dependency parsing or transformer models such as BERT. The system will employ models like those in Hugging Face Transformers or spaCy to categorize sentiments as positive, negative, or neutral, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in advanced NLP techniques and contributes to applications in e-commerce and customer feedback analysis. Students will also address ethical considerations, ensuring unbiased sentiment detection and responsible handling of user-generated content. Dataset: SemEval-2014 Task 4 Dataset http://alt.qcri.org/semeval2014/task4/, Amazon Product Reviews Dataset (Kaggle) https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products

158	Multimodal Emotion Recognition from Text and Voice	This project aims to develop a multimodal system for emotion recognition by integrating speech and text features from audio recordings and their transcriptions, using machine learning models like transformers for text and convolutional neural networks (CNNs) for audio. Students will preprocess text data with tokenization and speech data by extracting features like MFCCs or spectrograms, utilizing libraries such as Hugging Face Transformers, librosa, and PyTorch. The system will classify emotions (e.g., happy, sad, angry) by combining modalities through techniques like feature fusion or attention mechanisms, with performance evaluated using metrics such as accuracy, precision, recall, and F1-score. The project provides hands-on experience in multimodal machine learning and contributes to applications in human-computer interaction and mental health monitoring. Students will also address ethical considerations, ensuring privacy in handling sensitive audio and text data and mitigating biases in emotion detection. Dataset: IEMOCAP Dataset https://sail.usc.edu/iemocap/, MELD (Multimodal EmotionLines Dataset) https://affective-meld.github.io/, CMU-MOSEI Dataset https://www.kaggle.com/datasets/gnurtqh/cmu-mosei
159	Zero-Shot Text Classification Using LLMs	This project focuses on developing a zero-shot text classification system using large language models (LLMs) like BERT, GPT, or T5 to categorize text into predefined classes without requiring task-specific training data. Students will design prompts to leverage LLMs' pre-trained knowledge, utilizing libraries such as Hugging Face Transformers to process inputs and generate predictions. The system will classify texts, such as reviews or social media posts, into categories like sentiment or topic, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in prompt engineering and zero-shot learning, applicable to scenarios with limited labeled data. Students will also explore ethical considerations, including mitigating biases in model predictions and ensuring robust performance across diverse datasets. Dataset: Sentiment140 Dataset (Kaggle) https://www.kaggle.com/datasets/kazanova/sentiment140, AG News Dataset https://www.kaggle.com/datasets/amananandrai/ag-news-classification-dataset

160	Intent Detection in Customer Service Chatbots	This project aims to develop a natural language processing (NLP) system to detect user intents in customer service chatbot interactions, such as refund requests, product inquiries, or technical support, using models like BERT or traditional classifiers like SVM. Students will preprocess customer query texts, handling noise like slang or typos, and extract features using techniques such as TF-IDF or transformer embeddings. The system will classify intents into predefined categories, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and intent recognition, contributing to improved chatbot functionality in customer service applications. Students will also address ethical considerations, ensuring unbiased intent detection and protecting user privacy in sensitive query data. Dataset: Banking77 Dataset https://huggingface.co/datasets/PolyAI/banking77, CLINC150 Dataset https://huggingface.co/datasets/DeepPavlov/clinc150, Customer Support Dataset (Kaggle) https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter
161	Automated Email Classification	In this project, a program is developed to automatically classify business emails into predefined categories (spam, customer complaints, customer queries, etc). Optionally, the program can also generate written justifications for the classification.
162	Spam Detection Using NLP	This project aims to develop a natural language processing (NLP) system to classify emails or SMS messages as spam or non-spam (ham) by leveraging text features and machine learning models like Naive Bayes, SVM, or transformer-based models such as BERT. Students will preprocess text data, cleaning noise like special characters, URLs, or emojis, and extract features using techniques such as TF-IDF, word embeddings, or contextual embeddings. The system will be trained to accurately distinguish spam from legitimate messages, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and text classification, contributing to applications in email filtering and cybersecurity. Students will also address ethical considerations, ensuring unbiased classification a Dataset: SMS Spam Collection Dataset https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset, Enron Email Dataset https://www.kaggle.com/datasets/wcukierski/enron-email-dataset

163	Event Extraction from News Articles	This project focuses on developing a natural language processing (NLP) system to extract structured events—entities and details such as "who" (participants), "what" (action), and "where" (location)—from unstructured news articles. Students will use techniques like named entity recognition (NER) and relation extraction, leveraging models such as BERT or spaCy for entity identification and dependency parsing for event structure. The system will be trained to output structured data (e.g., JSON) representing events, with performance evaluated using metrics like precision, recall, and F1-score for entity and event accuracy. The project provides hands-on experience in advanced NLP and information extraction, applicable to applications like automated journalism and event tracking. Students will also address ethical considerations, ensuring accurate representation of events and handling potentially sensitive news content responsibly Dataset: ACE 2005 Dataset https://catalog.ldc.upenn.edu/LDC2006T06, Event Detection Dataset (Kaggle) https://www.kaggle.com/c/events-detection
164	Question Generation from Text Passages	This project focuses on developing a natural language processing (NLP) system to automatically generate questions from text passages for educational or question-answering (QA) applications, utilizing generative models like T5, BART, or transformer-based architectures. Students will preprocess text passages to identify key information (e.g., entities, facts) using techniques like named entity recognition (NER) or dependency parsing, and fine-tune models to produce diverse question types (e.g., factual, inferential). The system will be implemented using libraries like Hugging Face Transformers, with performance evaluated through metrics such as BLEU, ROUGE, and human-judged question relevance and grammaticality. The project offers hands-on experience in NLP and question generation, contributing to educational tools and automated assessment systems. Students will also address ethical considerations, ensuring questions are unbiased, factually accurate, and suitable for diverse learners. Dataset: SQuAD (Stanford Question Answering Dataset) https://rajpurkar.github.io/SQuAD-explorer/, Question Generation Dataset (Hugging Face) https://huggingface.co/iarfmoose/t5-base-question-generator

165	Fake Review Detection on E-Commerce Platforms	This project aims to develop a natural language processing (NLP) system to detect inauthentic or fake reviews on e-commerce platforms by combining text analysis with metadata features, such as reviewer behavior or rating patterns. Students will preprocess review texts, handling noise like emojis or slang, and extract features using techniques like TF-IDF, word embeddings, or transformer models (e.g., BERT). Metadata, such as review frequency or account age, will be integrated to enhance detection. The system will employ machine learning models like SVM, Random Forest, or fine-tuned LLMs to classify reviews as authentic or fake, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and fraud detection, contributing to trust in e-commerce ecosystems. Students will also address ethical considerations, ensuring fairness in detection to avoid mislabeling genuine reviews and protecting user privacy. Dataset: Amazon Reviews Dataset (Kaggle) https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products, Yelp Review Dataset https://www.kaggle.com/datasets/yelp-dataset/yelp-dataset,
166	Dialogue Act Classification	This project aims to develop a natural language processing (NLP) system to classify dialogue acts—categories like question, request, or statement—in conversational texts, enhancing dialogue system understanding. Students will preprocess dialogue data from datasets like chat logs or meeting transcripts, handling contextual dependencies and informal language. Using models such as BERT, LSTM, or traditional classifiers like SVM, implemented via libraries like Hugging Face Transformers or scikit-learn, the system will categorize utterances based on linguistic features and context. Performance will be evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and dialogue systems, applicable to chatbots and virtual assistants. Students will also address ethical considerations, ensuring unbiased classification and protecting user privacy in conversational data. Dataset: Switchboard Dialogue Act Corpus https://catalog.ldc.upenn.edu/LDC97S62, MRDA (Meeting Recorder Dialogue Act) Corpus https://opendatalab.com/OpenDataLab/MRDA, DailyDialog Dataset https://www.kaggle.com/datasets/thedevastator/dailydialog-multi-turn-dialog-with-intention-and

167	Automatic Paraphrase Detection	This project focuses on developing a natural language processing (NLP) system to detect whether two sentences are paraphrases, i.e., convey the same meaning despite different wording, using models like BERT, RoBERTa, or traditional similarity metrics. Students will preprocess sentence pairs, handling variations in syntax and vocabulary, and extract features using techniques like sentence embeddings or cosine similarity. The system will be implemented using libraries such as Hugging Face Transformers or scikit-learn to classify pairs as paraphrases or non-paraphrases, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and semantic analysis, applicable to applications like plagiarism detection and question answering. Students will also address ethical considerations, ensuring unbiased detection and handling sensitive text data responsibly. Dataset: MRPC (Microsoft Research Paraphrase Corpus) https://www.kaggle.com/datasets/doctri/microsoft-research-paraphrase-corpus, PAWS (Paraphrase Adversaries from Word Scrambling) https://github.com/google-research-datasets/paws, Quora Question Pairs Dataset https://www.kaggle.com/c/quora-question-pairs
168	Automatic Text Simplification	This project aims to develop a natural language processing (NLP) system to automatically simplify complex text, making it more accessible for language learners, children, or individuals with low literacy. The system will use generative models like T5, BART, or rule-based methods to perform tasks such as lexical simplification, sentence splitting, and syntax simplification. Students will preprocess input texts, handling complex vocabulary and sentence structures, using libraries like Hugging Face Transformers or NLTK. Performance will be evaluated using metrics like FKGL (Flesch-Kincaid Grade Level), SARI, and human-judged readability. The project provides hands-on experience in NLP and text transformation, contributing to educational tools and inclusive communication. Students will also address ethical considerations, ensuring simplifications preserve meaning and avoid condescension or cultural bias. Dataset: Simple English Wikipedia Dataset https://www.kaggle.com/datasets/ffatty/plain-text-wikipedia-simpleenglish, ASSET (A Dataset for Tuning and Evaluation of Sentence Simplification) https://aclanthology.org/2020.acl-main.424/

169	Style Transfer in Text (e.g., formal to informal)	This project focuses on developing a natural language processing (NLP) system for text style transfer, enabling transformation between styles like formal to informal, academic to conversational, or positive to neutral tones. Students will use generative models like T5, BART, or sequence-to-sequence architectures, fine-tuned on paired or unpaired datasets, to perform style transformations while preserving content meaning. Text preprocessing will involve handling stylistic markers and vocabulary differences, utilizing libraries such as Hugging Face Transformers or spaCy. Performance will be evaluated using metrics like BLEU, style accuracy (via classifiers), and human-judged coherence. The project provides hands-on experience in NLP and text generation, applicable to applications like automated content editing and chatbot personalization. Students will also address ethical considerations, ensuring transformations avoid introducing biases or altering intended meaning. Dataset: GYAFC (Grammarly's Yahoo Answers Formality Corpus), StylePTB Dataset https://opendatalab.com/OpenDataLab/StylePTB
170	Product Description Generation for E-Commerce	This project focuses on developing a natural language processing (NLP) system to generate compelling and accurate product descriptions for e-commerce platforms based on minimal input attributes, such as product name, category, features, and price. Students will use generative models like T5, BART, or GPT-based architectures, implemented via libraries like Hugging Face Transformers, to create coherent and persuasive descriptions. The system will preprocess structured attribute data, transforming it into natural language while ensuring stylistic consistency with e-commerce standards. Performance will be evaluated using metrics like BLEU, ROUGE, and human-judged quality for informativeness and appeal. The project provides hands-on experience in NLP and text generation, applicable to automated content creation for online retail. Students will also address ethical considerations, ensuring descriptions are truthful, avoid exaggeration, and respect cultural sensitivities. Dataset: Amazon Product Data Dataset (Kaggle) https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products

171	AI for Classifying Humor Types in Text	This project focuses on developing a natural language processing (NLP) system to identify and classify various types of humor, such as sarcasm, puns, and satire, in text data from sources like social media, forums, or literature. Students will preprocess text to handle linguistic nuances, employing techniques like tokenization, part-of-speech tagging, and contextual embeddings. Models like BERT, RoBERTa, or traditional classifiers like SVM will be implemented using libraries such as Hugging Face Transformers or scikit-learn to categorize humor types. Performance will be evaluated using metrics like accuracy, precision, recall, and F1-score, with qualitative analysis for contextual appropriateness. The project provides hands-on experience in advanced NLP and sentiment analysis, applicable to content moderation and entertainment applications. Students will also address ethical considerations, ensuring culturally sensitive classification and avoiding misinterpretation of humor in diverse contexts. Dataset: Sarcasm Detection Dataset (Kaggle) https://www.kaggle.com/danofer/sarcasm, Pun Detection and Interpretation Dataset https://alt.qcri.org/semeval2017/task7/
172	Happiness Index from Social Media	This project aims to develop a natural language processing (NLP) system to estimate public happiness levels by analyzing sentiment and emotional tone in social media posts, such as tweets or Reddit comments. Students will preprocess text data, cleaning noise like hashtags, emojis, or URLs, and extract features using sentiment analysis and emotion detection models (e.g., BERT, VADER, or RoBERTa). The system will quantify happiness through a composite index based on positive/negative sentiment and emotions like joy or sadness, using libraries like Hugging Face Transformers or TextBlob. Performance will be evaluated using metrics like correlation with ground-truth surveys, precision, and recall for sentiment/emotion classification. The project provides hands-on experience in NLP and sentiment analysis, applicable to social trend monitoring and mental health studies. Students will also address ethical considerations, ensuring user privacy, avoiding bias in sentiment interpretation, and handling sensitive emotional content responsibly. Dataset: Sentiment140 Dataset (Kaggle) https://www.kaggle.com/datasets/kazanova/sentiment140, Reddit Comments Dataset (Pushshift) https://zenodo.org/records/3608135

173	Fake Job Post Detection	This project aims to develop a natural language processing (NLP) system to identify fraudulent or misleading job postings by analyzing textual content and metadata, such as job descriptions, requirements, and company details. Students will preprocess text data, handling noise like vague phrasing or exaggerated claims, and extract features using techniques like TF-IDF, word embeddings, or transformer models (e.g., BERT). Metadata, such as posting frequency or company verification status, will be integrated to improve detection accuracy. The system will employ machine learning models like Random Forest, SVM, or fine-tuned LLMs, implemented via libraries like scikit-learn or Hugging Face Transformers, to classify postings as legitimate or fraudulent. Performance will be evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in NLP and fraud detection, contributing to safer job search platforms. Students will also address ethical considerations, ensuring unbiased classification to avoid mislabeling legitimate postings and protecting user trust. Dataset: Fake Job Postings Dataset (Kaggle) https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction, Employment Scam Aegean Dataset (EMSCAD) https://www.kaggle.com/datasets/amruthjithrajvr/recruitment-scam
174	AI-generated Content Detection	This project involves creating a system to distinguish AI-generated text from human-written content by analyzing linguistic patterns, such as vocabulary diversity, sentence structure, and stylistic features, using machine learning classification models. Students will preprocess datasets of labeled texts, extracting features like n-grams, perplexity scores, or embeddings from models like BERT to train classifiers such as SVM, Random Forest, or neural networks. The system aims to improve detection accuracy in real-world applications, such as content moderation or academic integrity checks, while addressing challenges like evolving AI generation techniques. Performance will be evaluated using metrics like accuracy, precision, recall, and F1-score, with potential incorporation of ensemble methods for robustness. Students will also explore ethical implications, including the potential for false positives affecting human creators and the need for transparency in detection algorithms. Dataset: AI Vs Human Text Dataset (Kaggle) https://www.kaggle.com/datasets/shanegerami/ai-vs-human-text, AI Text Detection Pile Dataset (Hugging Face) https://huggingface.co/datasets/artem9k/ai-text-detection-pile

175	Essay Grading using NLP	This project aims to develop a natural language processing (NLP) system to automatically evaluate and grade student essays by assessing content relevance, structural coherence, and grammatical accuracy. Students will preprocess essay texts, extracting features like lexical diversity, coherence metrics, and grammar errors using tools such as spaCy, Hugging Face Transformers, or NLTK. The system will employ machine learning models, such as BERT for semantic analysis or regression models for scoring, to assign grades based on rubrics for content, organization, and language quality. Performance will be evaluated using metrics like mean squared error (MSE), correlation with human scores, and accuracy for grammar detection. The project provides hands-on experience in NLP, text classification, and regression, contributing to automated educational assessment tools. Students will also address ethical considerations, ensuring fair grading across diverse writing styles and mitigating biases in automated evaluation. Dataset: ASAP (Automated Student Assessment Prize) Dataset https://www.kaggle.com/competitions/asap-aes/data, Essay Evaluation Dataset (Hugging Face) https://huggingface.co/KevSun/IELTS_essay_scoring
176	Financial News Sentiment Analysis	Identification of sentiment in a publicly available Kaggle financial news collection using deep learning methods.
177	Sarcasm Detection in News Articles	Identification of sentiment in a news collection publicly available in Kaggle with focus on detecting sarcasm using deep learning methods.
178	Global News Sentiment Analysis	Identification of sentiment in a publicly available Kaggle global news collection using deep learning methods.
179	AG News Sentiment Analysis	Identification of sentiment in the Kaggle AG news sentiment classification dataset using deep learning methods.
180	Space News Sentiment Analysis	Identification of sentiment in the publicly available Kaggle space news collection using deep learning methods.

Theme: Speech, Audio & Music

#	Title	Description
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181	Age Estimation from Voice Signals	This project focuses on developing a machine learning (ML) system to estimate a speaker's age group (e.g., child, young adult, middle-aged, senior) from voice signals by analyzing acoustic features like pitch, formant frequencies, and spectral characteristics. Students will preprocess audio data, extracting features such as Mel-frequency cepstral coefficients (MFCCs), pitch contours, or spectrograms using libraries like librosa or Praat. Models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based architectures will be trained to classify age groups, implemented via frameworks like TensorFlow or PyTorch. Performance will be evaluated using metrics like accuracy, mean absolute error (MAE), and F1-score. The project provides hands-on experience in audio signal processing and ML, applicable to fields like human-computer interaction and demographic analysis. Students will also address ethical considerations, ensuring privacy in handling voice data and mitigating biases related to gender, accent, or language. Dataset: Mozilla Common Voice Dataset https://commonvoice.mozilla.org/en/datasets, TIMIT Acoustic-Phonetic Continuous Speech Corpus https://catalog.ldc.upenn.edu/LDC93S1
182	Language Identification from Audio	This project aims to develop a machine learning (ML) system to identify the spoken language in short audio clips by analyzing acoustic features and speech patterns. Students will preprocess audio data, extracting features like Mel-frequency cepstral coefficients (MFCCs), spectrograms, or phoneme distributions using libraries such as librosa or SpeechBrain. Models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based architectures will be trained to classify languages, implemented via frameworks like PyTorch or TensorFlow. Performance will be evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio signal processing and ML, applicable to multilingual speech recognition and language detection systems. Students will also address ethical considerations, ensuring robust performance across diverse accents and dialects and protecting privacy in handling audio data. Dataset: Mozilla Common Voice Dataset https://commonvoice.mozilla.org/en/datasets

183	Speaker Diarization using Audio Segmentation	This project focuses on developing a speaker diarization system to segment audio recordings and label segments by individual speakers, leveraging audio processing and machine learning techniques. Students will preprocess audio data, extracting features like MFCCs, spectrograms, or embeddings from models like x-vectors or ECAPA-TDNN, using libraries such as librosa, PyTorch, or SpeechBrain. The system will cluster or classify speech segments to identify "who spoke when," with performance evaluated using metrics like Diarization Error Rate (DER), accuracy, and purity. The project provides hands-on experience in audio signal processing and machine learning, applicable to applications like meeting transcription and multimedia analysis. Students will also address ethical considerations, ensuring privacy in handling audio data and mitigating biases in speaker identification. Dataset: VoxCeleb Dataset https://www.robots.ox.ac.uk/~vgg/data/voxceleb/, AMI Meeting Corpus https://groups.inf.ed.ac.uk/ami/corpus/
184	Sound Event Detection in Urban Environments	This project aims to develop a sound event detection (SED) system to identify and classify environmental audio events, such as sirens, car horns, or construction noises, in urban settings using audio processing and machine learning techniques. Students will preprocess audio data by extracting features like Mel-frequency cepstral coefficients (MFCCs), log-mel spectrograms, or embeddings from pre-trained models like VGGish or YAMNet, using libraries such as librosa, TensorFlow, or PyTorch. The system will employ models like convolutional neural networks (CNNs) or transformers to classify or localize sound events, with performance evaluated using metrics like F1-score, precision, recall, and error rate. The project provides hands-on experience in audio signal processing and deep learning, applicable to applications like smart city monitoring and public safety systems. Students will also address ethical considerations, ensuring privacy in audio data collection and mitigating biases in event detection. Dataset: UrbanSound8K Dataset https://urbansounddataset.weebly.com/urbansound8k.html, ESC-50 (Environmental Sound Classification) Dataset https://github.com/karolpiczak/ESC-50

185	Environmental Sound Classification	This project aims to develop a sound classification system to identify and categorize everyday environmental sounds, such as dog barks, rain, engine noise, or footsteps, using audio processing and machine learning techniques. Students will preprocess audio data by extracting features like Mel-frequency cepstral coefficients (MFCCs), log-mel spectrograms, or embeddings from pre-trained models like VGGish or YAMNet, utilizing libraries such as librosa, TensorFlow, or PyTorch. The system will employ models like convolutional neural networks (CNNs) or transformers to classify sounds into predefined categories, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio signal processing and deep learning, applicable to applications like smart home systems, surveillance, and environmental monitoring. Students will also address ethical considerations, ensuring privacy in audio data handling and mitigating biases in sound classification. Dataset: ESC-50 (Environmental Sound Classification) Dataset https://www.kaggle.com/datasets/mmoreaux/environmental-sound-classification-50, UrbanSound8K Dataset https://urbansounddataset.weebly.com/urbansound8k.html
186	Bird Species Identification from Song Recordings	This project aims to develop a machine learning system to identify bird species based on their vocalizations, leveraging audio processing and classification techniques. Students will preprocess bird song recordings by extracting features like Mel-frequency cepstral coefficients (MFCCs), spectrograms, or embeddings from pre-trained models like VGGish or BirdNET, using libraries such as librosa, TensorFlow, or PyTorch. The system will employ models like convolutional neural networks (CNNs) or transformers to classify audio clips into specific bird species, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio signal processing and deep learning, applicable to biodiversity monitoring and ecological research. Students will also address ethical considerations, ensuring responsible use of environmental data and minimizing disturbance to wildlife during data collection. Dataset: Xeno-Canto Dataset https://xeno-canto.org/, BirdCLEF Dataset (Kaggle) https://www.kaggle.com/c/birdclef-2023/data

187	Instrument Classification in Music Tracks	This project aims to develop a machine learning system to detect and classify musical instruments (e.g., piano, guitar, violin) in audio clips using audio signal processing and deep learning techniques. Students will preprocess audio data by extracting features like Mel-frequency cepstral coefficients (MFCCs), spectrograms, or embeddings from pre-trained models such as VGGish or OpenL3, utilizing libraries like librosa, TensorFlow, or PyTorch. The system will employ convolutional neural networks (CNNs) or transformers to classify instruments, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio processing and classification, applicable to music information retrieval, audio production, and automated music analysis. Students will also address ethical considerations, ensuring proper handling of copyrighted audio data and mitigating biases in instrument detection across diverse music genres. Dataset: IRMAS (Instrument Recognition in Musical Audio Signals) Dataset https://www.upf.edu/web/mtg/irmas, NSynth Dataset https://magenta.tensorflow.org/datasets/nsynth
188	Speech Enhancement in Noisy Environments	This project aims to develop a machine learning system for speech enhancement, focusing on removing background noise (e.g., traffic, crowd, or wind) from speech recordings to improve audio clarity. Students will preprocess audio data by extracting features like log-mel spectrograms or waveforms, using libraries such as librosa, PyTorch, or TensorFlow. The system will employ deep learning models, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based architectures like DeepSpeech or WaveNet, to denoise audio. Performance will be evaluated using metrics like Perceptual Evaluation of Speech Quality (PESQ), Signal-to-Noise Ratio (SNR), and human-judged clarity. The project provides hands-on experience in audio signal processing and deep learning, applicable to applications like telecommunication, voice assistants, and hearing aids. Students will also address ethical considerations, ensuring privacy in handling speech data and minimizing biases in noise reduction across diverse voices and environments. Dataset: Voice Bank + DEMAND Dataset https://datashare.ed.ac.uk/handle/10283/2791, NOISEX-92 Dataset https://github.com/speechdnn/Noises, CHiME-5 Dataset https://www.chimechallenge.org/challenges/chime5

189	Music Mood Classification	This project aims to develop a machine learning system to classify songs into mood categories (e.g., happy, sad, energetic, calm) based on their audio features, leveraging audio signal processing and deep learning techniques. Students will preprocess audio data by extracting features like Mel-frequency cepstral coefficients (MFCCs), chroma features, tempo, or embeddings from pre-trained models like VGGish or MusiCNN, using libraries such as librosa, TensorFlow, or PyTorch. The system will employ models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformers to classify songs, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio processing and emotion recognition, applicable to music recommendation systems and media analysis. Students will also address ethical considerations, ensuring fair representation of diverse music genres and handling copyrighted audio data responsibly. Dataset: EmoMusic Dataset https://cvml.unige.ch/databases/emoMusic/, MTG-Jamendo Dataset https://mtg.github.io/mtg-jamendo-dataset/
190	Emotion Recognition from Speech Audio	Use machine learning to implement a speech emotion recognition system. The project will employ the publicly available voice dataset recorded by professional actors (Ravee, Crema, Savee and Tess) to identify sadness, anger, disgust, fear, and happiness. [Data: https://www.kaggle.com/datasets/dmitrybabko/speech-emotion-recognition-en/data]
191	Music Genre Classification with Deep Learning	Implement a deep model for music genre classification. Train and test the model using publicly available music genre recognition datasets.

192	Singing Voice Evaluation System	This project aims to develop a machine learning system to assess the accuracy of a singing voice in following the melody of a reference song, focusing on pitch, timing, and note accuracy. Students will preprocess audio data by extracting features like pitch contours, Mel-frequency cepstral coefficients (MFCCs), or chroma features, using libraries such as librosa, PyTorch, or Essentia. The system will employ models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based architectures to compare the sung melody against a reference, potentially using dynamic time warping (DTW) or sequence alignment for evaluation. Performance will be measured using metrics like pitch accuracy, note error rate, and perceptual quality scores. The project provides hands-on experience in audio signal processing and machine learning, applicable to music education, karaoke apps, and vocal coaching tools. Students will also address ethical considerations, ensuring fair evaluation across diverse vocal styles and protecting user privacy in audio data. Dataset: VoxCeleb Dataset https://www.robots.ox.ac.uk/~vgg/data/voxceleb/, DAMP (Digital Archive of Mobile Performances) Dataset https://ccrma.stanford.edu/damp/
193	Audio-Based Depression Detection	This project aims to develop a machine learning system to detect signs of depression from vocal features in speech recordings, leveraging audio signal processing and deep learning techniques. Students will preprocess audio data by extracting features such as pitch, prosody, Mel-frequency cepstral coefficients (MFCCs), or embeddings from pre-trained models like VGGish or HuBERT, using libraries like librosa, PyTorch, or SpeechBrain. The system will employ models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformers to classify speech as indicative of depression or non-depression, with performance evaluated using metrics like accuracy, precision, recall, and F1-score. The project provides hands-on experience in audio processing and mental health applications, contributing to early detection tools for mental health care. Students will also address ethical considerations, ensuring strict privacy protection for sensitive audio data, mitigating biases across diverse voices, and avoiding stigmatization in mental health predictions. Dataset: DAIC-WOZ (Distress Analysis Interview Corpus - Wizard of Oz) https://dcapswoz.ict.usc.edu/,

194	Real-Time Voice Cloning	This project aims to develop a real-time voice cloning system that synthesizes a target speaker's voice from a small set of audio recordings, leveraging deep learning techniques for text-to-speech (TTS) synthesis. Students will preprocess audio data by extracting features like Mel-frequency cepstral coefficients (MFCCs), spectrograms, or speaker embeddings using models like VALL-E, Tacotron 2, or SV2TTS, implemented with libraries such as PyTorch, TensorFlow, or SpeechBrain. The system will use a small dataset of target voice samples to fine-tune a pre-trained model for voice cloning, aiming for natural-sounding output with minimal latency. Performance will be evaluated using metrics like Mean Opinion Score (MOS), similarity to the target voice, and inference speed. The project provides hands-on experience in audio processing, deep learning, and TTS, applicable to personalized voice assistants and media production. Students will address ethical considerations, including preventing misuse for deepfakes, ensuring consent for voice data, and protecting privacy. Dataset: VoxCeleb Dataset https://www.robots.ox.ac.uk/~vgg/data/voxceleb/, LibriTTS Dataset http://www.openslr.org/60/
195	Automatic Music Transcription	This project aims to develop a machine learning system for automatic music transcription (AMT), converting audio recordings of music into symbolic notation like MIDI, capturing elements such as pitch, onset, duration, and instrument. Students will preprocess audio data by extracting features like spectrograms, constant-Q transforms, or Mel-frequency cepstral coefficients (MFCCs), using libraries such as librosa, PyTorch, or TensorFlow. The system will employ deep learning models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based architectures (e.g., Onsets and Frames or Music Transformer) to predict note sequences. Performance will be evaluated using metrics like note-onset F1-score, polyphonic transcription accuracy, and MIDI alignment error. The project provides hands-on experience in audio signal processing and deep learning, applicable to music production, education, and analysis tools. Students will also address ethical considerations, ensuring proper handling of copyrighted music data and fair representation of diverse musical styles. Dataset: MAESTRO Dataset https://magenta.tensorflow.org/datasets/maestro, MusicNet Dataset https://homes.cs.washington.edu/~thickstn/musicnet.html

196	Spoken Language Fluency Scoring	This project aims to develop a system that evaluates the spoken language fluency of learners based on their speech recordings. Students will extract features such as speech rate, pause duration, and pronunciation accuracy using audio processing and machine learning techniques to generate fluency scores. [Data: https://commonvoice.mozilla.org/en/datasets .]
197	Laughter Detection in Audio	This project focuses on detecting and classifying laughter events from audio recordings using machine learning techniques. Students will extract acoustic features (e.g., MFCCs, pitch, energy) from labeled audio datasets and train classifiers to distinguish laughter from other sounds. [Data: Use public datasets such as https://research.google.com/audioset/dataset/laughter.html]
198	Song Detection using Machine Learning	Implement a deep learning model for recognising a song, and then train and test the model using publicly available song recognition datasets.
199	Accent Detection in Spoken English	This project aims to classify different English accents by extracting and analyzing audio features from speech recordings. Students will preprocess the audio data, apply feature extraction techniques (e.g., MFCCs), and train supervised learning models for classification. [Data: https://www.kaggle.com/datasets/rtatman/speech-accent-archive]
200	Gender Recognition from Voice	This project aims to identify a speaker's gender by analyzing voice characteristics such as pitch, tone, and frequency features. Students will process audio data, extract relevant features, and build machine learning models for accurate classification. [Data: https://www.kaggle.com/datasets/primaryobjects/voicegender]

Theme: Scheduling & Optimisation

#	Title	Description
201	Badminton Tournament Scheduling	In this project, an application is created for automated scheduling of a badminton tournament, given specific constraints, such as game format (round robin/cup system), availability of courts, etc. The program should aim to schedule matches so that fairness criteria will be fulfilled as accurately as possible (e.g., all the players have in average the same time to rest between the games etc.)
202	Football League Scheduling	In this project, an application is created for automated scheduling of a football league, given specific constraints, such as game format, available dates for teams, etc. The program should aim to schedule matches so that fairness criteria will be fulfilled as accurately as possible (e.g., all the teams have in average the same time to rest between the matches etc.)

203	Locomotive Scheduling	In this project, an application is created for automated scheduling of locomotives to work on specific trains, given specific constraints, such as number of cars and locomotive capacity, route, etc.
204	Nursing Staff Shift Scheduling	In this project, an application is created for automated scheduling of work shifts, given specific constraints, such as personal preferences of the nursing staff members, required staff at different times according to their specialisations, etc. The program should aim to schedule courses so that fairness criteria will be fulfilled as accurately as possible (e.g., all the staff has enough time to rest between shifts, personal preferences followed as much as possible, etc.)
205	Restaurant Staff Shift Scheduling	In this project, an application is created for automated scheduling of work shifts in a restaurant, given specific constraints, such as personal preferences of the staff members, required number of staff at specific times, etc. The program should aim to schedule courses so that fairness criteria will be fulfilled as accurately as possible (e.g., all the staff has enough time to rest between shifts, personal preferences followed as much as possible, etc.)
206	University Course Scheduling	In this project, an application is created for automated scheduling of university classes, given specific constraints, such as classroom availability, classroom size, class size, teacher availability etc. The program should aim to schedule courses so that fairness criteria will be fulfilled as accurately as possible (e.g., teacher and student workload is not too heavily focused on same days, etc.)
207	Workforce Scheduling and Optimization	In this project, design a tool that automates workforce scheduling based on factors such as employee availability, skillsets, and workload requirements, optimizing shift assignments to meet business needs while minimizing overtime and staffing gaps.
208	Elevator Optimisation	In this project, a program is developed to optimise a group of elevators in a skyscraper so that the total waiting time is minimised. The program should be tested with different configurations in a simulated traffic environment, and the program should consider fairness criteria so that all the users will be served within a reasonable time.
209	Marine Traffic Optimisation	In this project, an online (or offline) platform is developed for assisting the user to predict and optimise marine traffic. Public datasets will be used for simulating marine traffic.
210	Traffic Light Optimisation	In this project, a program is developed to optimise the traffic light cycles so that the total travel time is optimised. The program should be tested with different configurations in a simulated traffic environment, and the program should consider fairness criteria so that all the cars will reach their destination within a reasonable time.

211	Fleet Route Management and Optimization	In this project, design a platform that optimizes the routing and scheduling of vehicles for transportation or delivery services, considering factors such as traffic conditions, vehicle capacities, and delivery deadlines.
212	Route Finder Assistant for Flights	In this project, an online (or offline) platform is developed for assisting the user to find the optimal flight connections from the the origin to the destination airport. The flight connections can be modeled as a graph, and it can be based on a real flight network system or an imaginary one.
213	Route Finder Assistant for Subway	In this project, an online (or offline) platform is developed for assisting the user to find the optimal (fastest) route from the the origin to the destination station in a subway system. The subway system is modeled as a graph, and it can be based on a real subway system or an imaginary one.
214	Route Finder Assistant for Trains and Buses	In this project, an online (or offline) platform is developed for assisting the user to find the optimal route from the the origin to the destination station in a network containing train and bus lines. The user can search for the optimal route using different criteria, such as the fastest or the cheapest route. The system can be modeled as a graph, based on real or imaginary railway and bus connections.
215	Airline Crew Scheduling	This project aims to develop an optimization system to assign flight crews efficiently while complying with aviation regulations, crew qualifications, and availability constraints. Students will model the scheduling problem, apply optimization or heuristic algorithms, and evaluate the solution's efficiency. [Data: https://www.kaggle.com/datasets/ramirobentes/flights-in-brazil]
216	Airport Runway Scheduling	This project aims to develop an optimization model to allocate takeoff and landing slots at airports to minimize overall flight delays, considering factors such as runway availability, air traffic, and weather conditions. Students will design and implement algorithms to solve the scheduling problem and evaluate their performance. [Data: https://www.kaggle.com/datasets/ramirobentes/flights-in-brazil]
217	Hotel Room booking prediction	This project aims to develop a predictive model that forecasts hotel room bookings based on factors such as booking date, lead time, customer type, seasonality, and special events. Students will clean and preprocess the data, perform feature engineering, and apply machine learning algorithms to improve booking predictions and help hotels optimize room availability. [Data: https://www.kaggle.com/datasets/mojtaba142/hotel-booking]

218	Smart Classroom Timetable Generator	This project aims to build an AI-powered system that generates optimized school or classroom timetables by considering teacher availability, subject requirements, and room capacity. Students will design algorithms to automatically resolve scheduling conflicts and improve timetable efficiency. [Data: https://www.kaggle.com/datasets/smrezwanulazad/exam-schedule?]
219	Warehouse Staff Task Scheduling	This project aims to develop a system that dynamically schedules warehouse tasks and allocates staff based on workload, priorities, and shift availability. Students will design algorithms to improve operational efficiency, reduce idle time, and adapt to changing demands in real time. [Data: https://www.kaggle.com/datasets/prachi13/customer-analytics]
220	Delivery Drone Route Planning	This project aims to design an optimized route planning system for delivery drones by considering factors such as battery capacity, payload weight, distance, and delivery priorities. Students will implement algorithms to minimize travel time and energy consumption while ensuring all deliveries are completed within operational constraints. [Data: https://data.mendeley.com/datasets/m5vmmkvx22/5]
221	Prediction of Surgery Duration	This project aims to build a predictive model to estimate the duration of surgeries based on factors such as patient information, surgery type, surgeon experience, and medical history. Students will preprocess the data, extract relevant features, and apply machine learning algorithms to improve prediction accuracy and assist in operating room scheduling. [Data: https://www.kaggle.com/competitions/prediction-of-surgery-duration/overview]
222	On-Demand Driver Dispatch Optimization	This project aims to build a system that efficiently matches drivers to ride requests in real-time, minimizing passenger wait times and driver idle times. Students will explore optimization algorithms or reinforcement learning to dynamically assign rides based on location, ETA, and demand patterns. [Data: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page .]
223	Garbage Truck Routing and Scheduling	This project aims to design an optimized routing and scheduling system for garbage trucks to minimize fuel consumption, reduce operational costs, and ensure timely waste collection. Students will consider factors such as route distance, traffic conditions, truck capacity, and collection frequency, and implement algorithms to improve efficiency. [Data: https://www.kaggle.com/datasets/lsind18/solid-waste-and-recycling-collection-routes]

224	Stadium Event Scheduling Assistant	This project aims to develop a scheduling system that coordinates multiple events in sports venues by considering factors such as event type, duration, setup/teardown time, and venue availability. Students will design algorithms to avoid scheduling conflicts, optimize resource usage, and maximize venue occupancy. [Data: https://www.kaggle.com/datasets/antimoni/football-stadiums]
225	Factory Production Line Scheduling	This project aims to develop a scheduling system that efficiently manages job assignments on manufacturing lines by considering factors such as job priority, machine availability, and processing time. Students will implement optimization algorithms to minimize production delays and maximize throughput. [Data: https://www.kaggle.com/datasets/ziya07/manufacturing-production-data]

Theme: Business & Management Systems

#	Title	Description
226	Student Accommodation Management System	In this project, a database application and an online (or offline) user interface will be developed for managing student dormitory accommodation. The user interface will allow users with different privileges to search and reserve accommodation for specific periods of time. The application should support dormitories with different number of beds.
227	Supermarket Stock Management System	In this project, a database application and a user interface will be developed for managing the supermarket stock. The user interface will allow users with different privileges to search, add, reserve, remove items from the stock. The application should also keep track on the expiration dates of the food items to notify them to be discarded on time.
228	Warehouse Stock Management System	In this project, a database application and a user interface will be developed for managing the warehouse stock. The user interface will allow users with different privileges to search, add, reserve, remove items from the stock. The application should also keep track on the location of the items in the warehouse.
229	Farm Cargo Management System	This project involves developing a system to manage the logistics of transporting agricultural goods from farms to markets. Students will implement modules for cargo scheduling, route planning, vehicle tracking, and inventory monitoring to ensure timely and efficient delivery of produce. [Data: https://github.com/topics/cargo-management-software .]

230	Customer Relationship Management System for a Major Hotel Chain	This project focuses on building a CRM system tailored for the hospitality industry to track guest preferences, stay history, and engagement metrics. Students will implement features such as personalized service recommendations, loyalty tracking, and feedback management to enhance guest satisfaction and retention. [Data: https://github.com/aliasar1/Hotel-Management-System .]
231	Business-to-Business Relationship Management System	This project aims to develop a system for managing interactions, communications, and contracts between a large corporation and its corporate clients. Students will implement modules for client profiles, contract lifecycle tracking, meeting logs, and analytics to support strategic decision-making in B2B relationships. [Data: https://github.com/kamgrzeski/SoftCRM .]
232	Enterprise Resource Management System for a Major Logistics Firm	This project involves developing an integrated platform to manage core operations of a logistics firm, including supply chain coordination, workforce scheduling, and fleet tracking. Students will design modules for inventory control, delivery route optimization, staff management, and real-time monitoring of logistics activities. [Data: https://github.com/strongloop/acme-freight .]
233	Attendance Recording System for a University Class	This project aims to develop a system that automates attendance tracking and reporting for university classes. Students will implement features such as student check-in via QR codes or RFID, attendance analytics, and automated report generation for instructors and administrators. [Data: https://github.com/msaf9/student-attendance-management-system .]
234	Checkout System for a Major Supermarket Chain	This project involves developing a digital point-of-sale (POS) and billing system for a large supermarket chain. Students will implement features such as barcode scanning, inventory lookup, dynamic pricing, receipt generation, and integration with payment gateways. The system should support real-time updates and transaction logging. [Data: https://github.com/TomGiz/supermarket_checkout]
235	Supply Chain Optimization	This project involves developing a Python-based supply chain optimization project to minimize costs by modeling inventory management, transportation routing, and production planning. It aims to use libraries like PuLP for linear programming, NetworkX for routing, and pandas for data analysis to optimize inventory levels, delivery routes, and production schedules. [Data: kaggle.com/c/inventory-optimization]

236	Anomaly Detection of Fraudulent Financial Transactions	This project aims to develop a predictive model using machine learning algorithms to accurately identify fraudulent financial transactions based on features such as purchase time, value, source, and ID. [Data: github.com/amazon-science/fraud-dataset-benchmark]
237	Predictive Analytics of Credit Card Applications	The project will use retrospective data of credit card applications to train a predictive model for classifying credit card applications. [Data: archive.ics.uci.edu/dataset/27/credit+approval]
238	Business Intelligence Dashboard for Retail Chains	This project focuses on developing interactive dashboards to monitor key business metrics such as sales performance, revenue trends, inventory levels, and customer behavior across multiple retail locations. Students will use data visualization tools to help stakeholders make informed, data-driven decisions. [Data: https://github.com/TheMrityunjayPathak/RetailStoreSalesDashboard .]
239	Vendor Rating System for Procurement	This project involves building a system to evaluate and rate suppliers based on performance metrics such as product quality, delivery timeliness, cost-effectiveness, and reliability. Students will design scoring algorithms, visualize vendor performance trends, and support decision-making in procurement processes. [Data: https://github.com/Darshan1904/Sparkathon .]
240	Human Resource Management System for a Tech Firm	This project focuses on developing a comprehensive HR management system tailored for a tech firm, automating key processes such as employee hiring, payroll management, and performance evaluations. Students will implement modules for applicant tracking, salary processing, leave management, and periodic review generation. [Data: https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset .]
241	Customer Feedback and Analytics Platform	This project involves building a platform to collect, analyze, and visualize customer feedback from surveys, reviews, or support interactions. Students will apply natural language processing (NLP) techniques to extract sentiment, key topics, and trends, providing actionable insights for business improvement. [Data: https://www.kaggle.com/datasets/bittlingmayer/amazonreviews .]

242	Retail Shelf Management System	This project aims to develop a system that optimizes product placement and shelf replenishment in retail stores. Students will design algorithms to improve shelf space utilization, ensure stock availability, and track product movement using sales data or sensor input. Visualization tools for planograms and restocking alerts will also be implemented. [Data: https://github.com/rokopi-byte/shelf_management .]
243	Customer Loyalty and Rewards Program System	This project involves developing a system to manage customer loyalty programs, including tracking reward points, managing membership tiers, and generating personalized offers based on customer behavior and purchase history. Students will design modules for point redemption, offer recommendations, and customer engagement analytics. [Data: https://github.com/PrathamBhatTech/Customer-Loyalty-Software]
244	Digital Invoice Management System	This project aims to build a system that streamlines the creation, tracking, and payment of digital invoices for businesses. Students will implement features such as automated invoice generation, payment status monitoring, due date reminders, and integration with client records and accounting tools. [Data: https://www.kaggle.com/datasets/cankatsrc/invoices .]
245	Business Travel and Expense Management System	This project focuses on developing a system to streamline the submission, approval, and tracking of employee travel requests and expense claims. Students will implement features such as travel itinerary logging, receipt uploads, automated approval workflows, and expense reporting for finance teams. [Data: https://github.com/topics/expense-management-system]
246	Banking Services Chatbot	In this project, an online platform is implemented with a chatbot for assisting the user in banking services.
247	Hotel Reservation Chatbot	In this project, an online platform is implemented with a chatbot for assisting the user to reserve hotel rooms, cancel bookings, etc.
248	Internet Shop Chatbot	In this project, an online platform is implemented with a chatbot for assisting the user to order products from an internet shop, ask for information about the products, etc.
249	Movie Theatre Chatbot	In this project, an online platform is implemented with a chatbot for assisting the user to buy movie tickets, ask questions about the movies in the theatre, etc.
250	Restaurant Reservation Chatbot	In this project, an online platform is implemented with a chatbot for assisting the user to book a table in a restaurant, cancel a booking, ask questions about the menu, etc.

251	Hotel Recommendation System	In this project, an online (or offline) platform will be developed to provide users with recommendations for hotels, based input from the users of the system. The platform allows users to rate different hotels and guesthouses and stores the ratings in a database, and the users can search for personally recommended hotels based on their ratings and the ratings by other users.
252	Movie Recommendation System	In this project, an online (or offline) platform will be developed to provide users with recommendations for movies, based input from the users of the system. The platform allows users to rate different movies and stores the ratings in a database, and the users can search for personally recommended movies based on their ratings and the ratings by other users.
253	Music Recommendation System	In this project, an online (or offline) platform will be developed to provide users with recommendations for songs or artists, based input from the users of the system. The platform allows users to rate different songs or artists and stores the ratings in a database, and then users can search for personally recommended songs or artists based on their ratings and the ratings by other users.
254	Restaurant Recommendation System	In this project, an online (or offline) platform will be developed to provide users with recommendations for restaurants, based input from the users of the system. The platform allows users to rate different restaurants and stores the ratings in a database, and then users can search for personally recommended restaurants based on their ratings and the ratings by other users.
255	Tourist Attraction Recommendation System	In this project, an online (or offline) platform will be developed to provide users with recommendations for tourist attractions, based input from the users of the system. The platform allows users to rate different attractions and stores the ratings in a database, and then users can search for personally recommended restaurants based on their ratings and the ratings by other users.
256	Freelancer Project Bidding and Management Platform	This project aims to develop a web-based platform where freelancers can browse available projects, submit bids, manage assigned tasks, and communicate with clients in a centralized workspace. Students will implement features such as project listings, bidding workflows, messaging, milestone tracking, and user profiles.
257	Corporate Asset Tracking System	This project involves building a system to monitor physical assets across departments or locations, including features for asset registration, location tracking, maintenance logging, and depreciation calculation over time. Students will design a dashboard for real-time status updates and generate reports for audits and financial planning. [Data: https://github.com/Pavel-Khan17/Asset-Tracking-System.]

258	Client Proposal and Contract Lifecycle Management System	This project focuses on building a system to manage the full lifecycle of client proposals and contracts, including creation, approval workflows, version control, and status tracking. Students will design features to support collaborative editing, deadline reminders, and secure document storage. [Data: https://www.kaggle.com/datasets/konradb/atticus-open-contract-dataset-aok-beta .]
259	Business Risk Assessment and Compliance System	This project aims to develop a system that evaluates business risks and monitors regulatory compliance across departments or projects. Students will design modules to identify potential financial, operational, or legal risks, assign severity scores, and generate compliance reports based on predefined rules or standards. [Data: https://www.kaggle.com/datasets/ka66ledata/project-management-risk-raw .]
260	Startup Investment and Portfolio Management System	This project involves building a system to manage and analyze startup investments, tracking funding rounds, key startup metrics (e.g., valuation, revenue, burn rate), and return on investment (ROI) for venture capital firms or incubators. Students will design a dashboard for portfolio monitoring and implement analytics to support investment decisions. [Data: https://www.kaggle.com/datasets/justinas/startup-investments .]

Theme: Game Bots

#	Title	Description
261	Game Bot for Captain's Mistress	In this project, a program is developed to play against a human opponent (or another algorithm) in the game of Captain's Mistress (https://www.mastersofgames.com/rules/captains-mistress-rules.htm). The bot can be based on hand-written rules or machine learning.
262	Game Bot for Checkers	In this project, a program is developed to play against a human opponent (or another algorithm) in the famous game of Checkers (https://www.mastersofgames.com/rules/draughts-rules.htm). The bot can be based on hand-written rules or machine learning.
263	Game Bot for Chinese Checkers	In this project, a program is developed to play against a human opponent (or another algorithm) in Chinese Checkers (https://www.mastersofgames.com/rules/chinese-checkers-rules.htm). The bot can be based on hand-written rules or machine learning, and it is enough to support the two-player version of the game.

264	Game Bot for Hnefatafl	In this project, a program is developed to play against a human opponent (or another algorithm) in the ancient game of Hnefatafl (http://aagenielsen.dk/simple_hnefatafl_rules.php). The bot can be based on hand-written rules or machine learning.
265	Game Bot for Fangqi	In this project, a program is developed to play against a human opponent (or another algorithm) in Fangqi (https://boardgamegeek.com/boardgame/42225/fangqi). The bot can be based on hand-written rules or machine learning.
266	Game Bot for Nine Men's Morris	In this project, a program is developed to play against a human opponent (or another algorithm) in Nine Men's Morris (https://www.mastersofgames.com/rules/morris-rules.htm). The bot can be based on hand-written rules or machine learning.
267	Game Bot for Five-in-a-Row	In this project, a program is developed to play against a human opponent (or another algorithm) in the famous game of Five-in-a-Row, an infinite variant of Tic-Tac-Toe (https://www.mathsisfun.com/games/gomoku.html). The bot can be based on hand-written rules or machine learning, and it could support different variants of the game.
268	Game Bot Tournament Framework	This project involves designing and implementing a tournament platform where multiple AI bots can compete across classic strategy games such as Tic-Tac-Toe, Connect Four, or simplified Chess. Students will build a modular framework to support game integration, bot registration, match scheduling, and leaderboard scoring. [Data: Use open-source game implementations, for example, please read reference: https://arxiv.org/pdf/2103.02943 .]
269	Generalized Game Bot using Rule Files	This project aims to build a game-playing bot that can read formal rule descriptions—such as those written in Game Description Language (GDL) or structured JSON—and learn to play various games without manual coding for each one. Students will design a rule parser, generate internal game logic, and implement planning or learning-based agents to play different games. [Data: Use rule sets from the General Game Playing Repository https://github.com/ggp-org/ggp-base .]
270	Zero-Knowledge Game Learning Agent	This project involves training an AI agent to learn and play a game without prior knowledge of its rules, relying solely on environment feedback and rewards. Students will implement reinforcement learning algorithms such as Deep Q-Networks (DQN) or Proximal Policy Optimization (PPO) to enable the agent to adapt and improve performance through trial and error. [Data: Use custom or existing environments from https://gymnasium.farama.org/]

271	AI Agent for Snake Game	This project aims to train an AI agent using reinforcement learning techniques (e.g., Q-learning or Deep Q-Networks) to play the classic Snake game. The goal is to enable the agent to navigate efficiently, avoid collisions, and maximize the score by collecting food. [Data: Use open-source environments such as https://github.com/grantsrb/Gym-Snake]
272	AI Agent for Tetris	This project focuses on developing an AI agent to play the classic game Tetris using either hand-crafted heuristics or deep Q-learning. Students will design and train the agent to place blocks efficiently, maximize score, and survive as long as possible in increasingly difficult game scenarios. [Data: Use open-source environments such as https://github.com/Kautenja/gym-tetris]
273	AI Agent for 2048 Puzzle	This project involves building an AI agent to play the 2048 game intelligently using algorithms like expectimax search or deep reinforcement learning. Students will implement game simulations, train or tune the agent, and evaluate its performance based on achieved scores and decision efficiency. [Data: Game environment can be implemented or adapted from open-source versions such as https://github.com/yangshun/2048-python]
274	Pac-Man Game Bot	This project aims to develop an intelligent agent to play the Pac-Man game by collecting pellets while avoiding ghosts. Students can explore classic search algorithms (e.g., A*, BFS) or train reinforcement learning models to optimize the bot's performance in different game scenarios. [Data: https://github.com/xujiachang1024/MDP-Pac-Man]
275	Obstacle-Avoiding Bot for 2D Grid Game	This project focuses on designing and training an intelligent agent to navigate a 2D grid-based game environment filled with dynamic obstacles. Students will implement pathfinding or reinforcement learning algorithms to enable the bot to reach its goal while avoiding collisions. Visualization and performance evaluation will be part of the final deliverables. [Data: https://github.com/ravasconcelos/rl_obstacle_avoidance]

Theme: AI for Healthcare

#	Title	Description
276	Breast Cancer Survival Analysis Assessing Gene Expression Impact	Analyze the impact of specific gene expressions on survival rates among breast cancer patients to identify potential prognostic biomarkers. [Data/Source: https://kmplot.com/analysis/]
277	Ovarian Cancer Prognostics Gene Expression Profiles and Patient Outcomes	This study focuses on ovarian cancer, correlating gene expression profiles with survival outcomes to enhance prognostic accuracy. [Data/Source: https://kmplot.com/analysis/]

278	Gene Expression and Survival Analysis of Lung Cancer Patients	Determine how different gene expressions influence survival probabilities in lung cancer patients, aiming to improve treatment strategies. [Data/Source: https://kmplot.com/analysis/]
279	Gastric Cancer Biomarker Validation via Survival Analysis	Leveraging survival data analysis, this project validates the relevance of biomarkers in gastric cancer, contributing to targeted therapy developments.
280	Analysis of Prognostic Value of Gene Expression in Colon Cancer	Investigate the prognostic value of gene expression levels in colon cancer, potentially leading to more personalized treatment plans. [Data/Source: https://kmplot.com/analysis/]
281	Prognostic Insights into the CALD1 Gene	Investigate the CALD1 gene's role and its prognostic value in various diseases, aiming to understand how its expression correlates with disease outcomes. [Source: https://doi.org/10.1201/9781003251903-8/]
282	Deciphering PROM1/PROM2 gene's Prognostic Significance	Analyze the prognostic significance of PROM1 and PROM2 genes in certain diseases, potentially leading to new insights into their roles in disease progression and patient outcomes. [Source: https://www.nature.com/articles/s41417-019-0109-7]
283	Analysing Proteomics Dataset using Machine Learning Based Data Clustering Approaches	Use Human Leukocyte Antigen (HLA) sequence dataset and apply unsupervised machine learning approaches (such as K-means, Gaussian Mixture Model (GMM)) to get any inherent clusters and/or subclusters within a set of alleles of HLA. The processing of the data will also involve understanding sequence alignment and preparing data ready for use by machine learning models. [Data: https://github.com/ANHIG/IMGTHLA/tree/Latest]
284	Analysing Proteomics Dataset using Machine Learning Based Data Projection Techniques	Use Human Leukocyte Antigen (HLA) sequence dataset and visually observe a group of alleles of HLA Protein Sequences by projecting high-dimensional sequence descriptors to a low-dimensional (2-D) space using techniques such as Principal Component Analysis (PCA), t-Stochastics Neighbourhood Embedding (t-SNE). The processing of the data will also involve understanding sequence alignment and preparing data ready for use by machine learning models. [Data: https://github.com/ANHIG/IMGTHLA/tree/Latest]
285	Predictive Analytics of Cervical Cancer Patients	Use retrospective data of cervical cancer patients risk factors to train a machine learning and then use this trained model to enable predictions for patients with/without cervical cancer. [Data: https://archive.ics.uci.edu/dataset/383/cervical+cancer+risk+factors]
286	AI-Based Risk Stratification of Pancreatic Cancer Patients	Create a risk scoring system using machine learning on clinical and genomic datasets to stratify pancreatic cancer prognosis. The project will involve data preprocessing, feature selection, and training classification or regression models to predict patient risk levels. Data: https://portal.gdc.cancer.gov/projects/TCGA-PAAD]

287	Sleep Disorder Classification Using Wearable Device Data	Analyze actigraphy and heart rate data to identify sleep disorders such as insomnia or sleep apnea. The project will involve extracting relevant features from time-series data collected by wearable devices and training machine learning models to classify different types of sleep disorders. [Data: https://physionet.org/content/sleep-edfx/1.0.0/]
288	Predicting Risk of Cardiovascular Events from EHR Data	Use clinical and lab data to predict the likelihood of heart attacks or strokes in at-risk patients. The project will involve cleaning and analyzing electronic health record (EHR) data, selecting key risk factors (e.g., blood pressure, cholesterol levels, smoking history), and building machine learning models to estimate cardiovascular risk.[Data: https://physionet.org/content/mimiciv/3.1/]
289	Predictive Analytics for Patient Readmission	Build a machine learning model to predict the likelihood of patient readmission within a certain timeframe. The project will involve analyzing hospital admission records, identifying key predictors (e.g., diagnosis, length of stay, prior admissions), and training classification models to support hospital decision-making and reduce unnecessary readmissions.[Data: https://physionet.org/content/mimiciii/1.4/]
290	AI for Mental Health Monitoring	Investigate the use of AI to detect signs of mental health conditions like depression. This project will involve analysing data such as text responses, voice recordings, or questionnaire results to extract behavioural or linguistic patterns, and training machine learning models to identify potential indicators of mental health issues. [Data: https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch]
291	Predicting Disease Outbreaks or Public Health Trends	Develop a machine learning model to predict the outbreak or spread of infectious diseases or forecast public health trends like vaccination rates. The project will involve analysing historical health records, environmental factors, and population mobility data to identify patterns and train predictive models that support early warning and public health planning.[Data: https://www.google.com/covid19/mobility/]
292	Diabetes Prediction using Classifier Algorithms	This project is about using machine learning techniques, likely classification algorithms, to predict the occurrence or classification of diabetes based on certain input features or variables. [Data: kaggle.com/code/adilashrafi/diabetes-prediction-accuracy-99]
293	Machine-learning-based Heart Disease Risk Factors Analysis	This project analyzes heart disease dataset to uncover insights into the distribution, correlations, and patterns within the data. The aim is to gain a deeper understanding of the dataset's characteristics, which can inform the development of predictive models and aid in identifying risk factors for heart disease. [Data: kaggle.com/code/georgyzubkov/heart-disease-exploratory-data-analysis/notebook]

294	Exploratory Data Analysis(EDA) on Diabetes Data Set	Exploratory Data Analysis (EDA) on the Diabetes dataset involves analyzing and visualizing various attributes to gain insights into the data distribution, relationships between features, and potential patterns. [Data: kaggle.com/code/imtkaggleteam/diabetes-analysis/notebook]
295	Analysis of Diabetic retinopathy using SHAP Explainability	This project analyzes diabetic retinopathy using SHAP explainability, aiming to understand the influence of various features on the condition's severity. [Data: kaggle.com/code/dimitreoliveira/diabetic-retinopathy-shap-model-explainability/notebook]

Theme: Cybersecurity

#	Title	Description
296	Supervised Learning for Intrusion Detection System	This project aims to develop a robust Intrusion Detection System (IDS) using supervised learning algorithms, including Random Forest and Support Vector Machine (SVM). It focuses on analyzing network traffic data to accurately identify and classify potential cyber threats, enhancing security measures. [Source: kaggle.com/code/abhaymudgal/intrusion-detection-system/notebook]
297	ANN-Based Intrusion Detection System	This project leverages Artificial Neural Networks (ANN) to create an advanced Intrusion Detection System (IDS) capable of analyzing and identifying cyber threats within network traffic with high accuracy, ensuring enhanced cybersecurity posture. [Source: kaggle.com/code/abhaymudgal/intrusion-detection-system/notebook]
298	Deep Learning-Based Anomaly Detection in Network Traffic	This project aims to apply deep learning models, such as Convolutional Neural Networks (CNNs) or Long Short-Term Memory networks (LSTMs), to detect anomalies and zero-day attacks in network traffic data. Students will process flow-based features, train deep models to identify deviations from normal behavior, and evaluate detection performance on benchmark datasets. [Data: https://www.kaggle.com/datasets/ziya07/network-traffic-anomaly-detection-dataset]
299	Transfer Learning for Cross-Network Intrusion Detection	This project explores how to apply transfer learning techniques to adapt intrusion detection models trained on one network dataset (e.g., NSL-KDD) to perform effectively on another with different characteristics (e.g., CICIDS2017). Students will implement and evaluate domain adaptation methods to improve generalization across datasets. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]

300	Supervised Malware Classification using Static and Dynamic Features	This project aims to train machine learning models to classify malware by leveraging both static features (such as file signatures and imported functions) and dynamic features (such as system calls and execution behavior). Students will explore feature extraction techniques and evaluate classifiers for effective detection. [Data: https://www.kaggle.com/c/malware-classification/data]
301	Lightweight IDS for IoT Devices using Decision Trees	This project aims to develop a lightweight intrusion detection system (IDS) using decision tree-based algorithms that can operate efficiently on IoT or edge devices with limited computational resources. Students will focus on model simplicity, low memory usage, and real-time performance. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]
302	Evaluation of IDS Performance on Imbalanced Datasets	This project aims to assess the performance of intrusion detection systems (IDS) when dealing with imbalanced cybersecurity datasets. Students will apply resampling techniques such as SMOTE, ADASYN, or class weighting to improve detection of minority attack classes and evaluate the impact on model accuracy and recall. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]
303	Explainable AI for Cyber Attack Classification	This project aims to apply explainable AI (XAI) techniques such as SHAP and LIME to interpret the decisions of intrusion detection systems (IDS). Students will train classification models on cyberattack data, then use explainability tools to identify key features influencing predictions and enhance model transparency. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]
304	Federated Learning for Distributed Intrusion Detection	This project aims to implement a federated learning framework for collaboratively training intrusion detection system (IDS) models across distributed network nodes, without exchanging raw data. Students will simulate data decentralization, design aggregation strategies, and evaluate detection performance under privacy constraints. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]
305	RNN-Based Intrusion Detection for Temporal Threat Patterns	This project aims to use recurrent neural networks (RNNs) to detect network intrusions by modeling temporal patterns in time-series traffic data. Students will preprocess network logs, extract sequential features, and train RNN-based models to identify abnormal behaviors and cyberattacks. [Data: https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection]